MODULI STACKS OF HYPERELLIPTIC CURVES

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ABSTRACT

MODULI STACKS OF HYPERELLIPTIC CURVES

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This thesis presents a comprehensive survey on the construction of $\overline{\mathcal{H}_g}$, the compactified moduli stack of hyperelliptic curves of genus $g \geq 2$. We utilize the Geometric Invariant Theory (GIT). The construction of moduli space of stable curves $\mathcal{M}_g$ is a classical result of the works of Deligne-Mumford and/or Gieseker, and in this work we restricted this machinery to the hyperelliptic locus. We started our construction through the Hilbert Schemes and their ν-canonical embeddings, then analyzed the stability conditions. Finally, we constructed the GIT quotient and showed it has finite and reduced stabilizers, upgrading it to a Deligne-Mumford stack and establish the equivalence between this Deligne-Mumford stack and the moduli functor of stable hyperelliptic curves.

Keywords: Moduli Stacks, Hyperelliptic Curves, Stable Curves, Geometric Invariant Theory, Deligne-Mumford Stacks

ÖZ

HİPERELİPTİK EĞRİLERİN MODULİ YIĞINLARI

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Bu tez, $g \geq 2$ cinsli hipereliptik eğrilerin kompaktlaştırılmış moduli yığını olan $\overline{\mathcal{H}_g}$'nin inşası üzerine kapsamlı bir inceleme sunmaktadır. Çalışmamızda Geometrik Değişmezler Teorisi (GIT) kullanılmaktadır. Stabil eğrilerin moduli uzayı olan $\mathcal{M}_g$'nin inşası, Deligne-Mumford ve/veya Gieseker'in çalışmalarının klasik bir sonucudur; biz bu mekanizmayı hipereliptik lokusa kısıtlayacağız. İnşamıza Hilbert Şemaları ve onların ν-kanonik gömmeleri aracılığıyla başlayıp, ardından stabilite koşullarını analiz edeceğiz. Son olarak, GIT bölümünü inşa edip sonlu ve indirgenmiş stabilizatörlere sahip olduğunu göstererek onu bir Deligne-Mumford yığınına terfi ettireceğiz. Ayrıca, bu Deligne-Mumford yığını ile stabil hipereliptik eğriler moduli funktoru arasındaki denkliği kuracağız.

Anahtar Kelimeler: Moduli Yığınları, Hipereliptik Eğriler, Stabil Eğriler, Geometrik Değişmezler Teorisi, Deligne-Mumford Yığınları

To my family

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CHAPTER 1

INTRODUCTION

In this thesis we will construct the moduli stacks of hyperelliptic curves. Construction of moduli stacks requires some machinery of algebraic geometry such as algebraic groups, stacks, Hilbert schemes, Geometric Invariant Theory.

In Chapter 1, we will introduce the main concepts of moduli theory, main results from algebraic geometry and definition of hyperelliptic curves.

In Chapter 2, we will introduce the basic machinery of algebraic groups, sites, stacks, and Deligne-Mumford stacks.

In Chapter 3, we will introduce the Hilbert schemes, Geometric Invariant Theory(GIT), stability conditions of GIT, stable reduction theorem, and some deformation theory.

In Chapter 4, we will constrcut the moduli stack of hyperelliptic curves through GIT and show that it is a smooth, seperated, proper Deligne-Mumford stack.

1.1 History of Moduli Theory

Moduli theory is a classical theory of algebraic geometry which deals with classifying and studying geometric objects up to some equivalence. Moduli theory started

with Bernhard Riemann's work on studying the isomorphism classes of Riemann surfaces of genus $g \geq 2$ and finding that they depend on $3g-3$ complex parameters in his [Rie57] paper. His idea was to consider the curves associated to these Riemann surfaces as branched covers of $\mathbb{P}^1$, instead of as abstract smooth curves. This method later further studied by Hurwitz in his [Hur91] paper. In that paper, Hurwitz showed that the moduli space of genus g compactified Riemann surfaces is

$$Hur_{d,g} = \left\{ [C \to \mathbb{P}^1] \;\middle|\; \begin{array}{l} C \text{ is a smooth curve of genus } g \\ C \to \mathbb{P}^1 \text{ is a branched covering of degree } d \end{array} \right\}.$$

This moduli space is now called the *Hurwitz space*.

Further aspects of moduli theory were later studied by the Italian school of algebraic geometry, specifically by Enriques, Castelnuovo, and Severi. They mostly studied families of algebraic curves but lacked the technical apparatus from commutative algebra needed to formalize the parameter spaces.

Later, Teichmüller introduced the Teichmüller spaces which parametrize the space of complex structures on a Riemann surface, and this work established the classical moduli space as the quotient of the Teichmüller space by the mapping class group. This work mainly deals with complex analytic spaces, especially *orbifolds*.

In the 1960s, Grothendieck revolutionized algebraic geometry and consequently moduli theory. His introduction of the *functor of points* approach redefined the moduli problem and the concept of moduli spaces. One then defines the moduli problem as a moduli functor from Sch^{op} to $Sets$ (or $Groupoids$) instead of building Hurwitz and Teichmüller spaces pointwise.

Grothendieck also introduced the *Hilbert* and *Picard* schemes which we use extensively in this thesis. Hilbert schemes became one of the most used parameter spaces in moduli theory.

Mumford introduced *Geometric Invariant Theory* in 1965 in his [Mum65] book. This theory provides a way to construct coarse moduli spaces as quasi-projective varieties using group actions. It also enables us to use several different parameter spaces for moduli problems. Hilbert schemes, Chow schemes, and binary forms are examples of such parameter spaces.

In 1969, Deligne and Mumford introduced their landmark work [DM69]. In this work they introduced the concept of stable curves and compactified moduli spaces. This paper also introduced *stacks* as an extension of schemes. They realized that in order to compactify a moduli space, one should consider nodal curves with finite automorphisms along with smooth curves. This way, we can construct moduli stacks which is a more general concept of moduli space than the classical one.

The most famous usage of moduli spaces is arguably the Wiles' work on the proof of Fermat's Last Theorem, which relies heavily on coarse moduli spaces of elliptic curves.

Other important applications of moduli spaces include the Topological Modular Forms (TMF), Gromov-Witten Invariants, TQFT.

- TMF is a bridge between algebraic geometry and stable homotopy theory. It is constructed by taking the moduli stack of elliptic curves and equipping it with E_∞-ring spectra.

- Gromov-Witten invariants are defined as the intersection theory on the Deligne-Mumford compactification of the moduli space of curves with n-marked points $\overline{\mathcal{M}_{g,n}}$.

- The boundary data of the $\overline{\mathcal{M}_{g,n}}$ dictates the sewing axioms of TQFT.

1.2 Moduli Problem

A moduli problem is a problem of classifying some geometric objects. We have a collection of geometric objects and we want to classify them up to some equivalence.

Definition 1.2.0.1. A **moduli problem** is a collection $\mathcal{X}$ of objects and an equivalence relation $\sim$ on $\mathcal{X}$.

Definition 1.2.0.2. A **moduli functor** is a functor $\mathcal{M} \in Psh(Sch)$ given by

$$\mathcal{M}(T) = \{\text{families over } T\}/\sim_T, \quad \mathcal{M}(f : T' \to T) = f^* : \mathcal{M}(T) \to \mathcal{M}(T')$$

Definition 1.2.0.3. Let $\mathcal{M}$ be a moduli functor, then a scheme M is a **fine moduli space** of $\mathcal{M}$ if it represents $\mathcal{M}$, i.e. if it satisfies the condition $\mathcal{M}(T) \cong \mathrm{Hom}(T, M)$ for all T.

Unfortunately, such a fine moduli space does *not* always exist. When a fine moduli space does not exist, we try to find another type of moduli space.

Definition 1.2.0.4. A **coarse moduli space** of $\mathcal{M}$ is a scheme M and a natural transformation $\phi : \mathcal{M} \to \mathrm{Hom}(\cdot, M)$ such that

1. $\phi_{\mathrm{Spec}(k)} : \mathcal{M}(\mathrm{Spec}(k)) \to \mathrm{Hom}(\mathrm{Spec}(k), M)$ is bijective
2. For any scheme N and natural transformation $\psi : \mathcal{M} \to \mathrm{Hom}(\cdot, N)$, there exists a unique morphism of schemes $f : M \to N$ such that $\psi = h_f \circ \phi$, where $h_f : \mathrm{Hom}(\cdot, M) \to \mathrm{Hom}(\cdot, N)$ is the natural transformation of presheaves.

1.3 Main Theorems of Algebraic Curves

In this section, we will state some of the main theorems of algebraic curves which we will use extensively in this thesis. Since these theorems are well-established, we will not provide proofs for them. Curious readers may refer to [EH24, Ful08, Gri89, Har77].

Definition 1.3.0.1. Let C be a smooth projective curve. A **divisor** D of C is a formal sum of points on C such as $D = \sum_{P \in C} n_P P$ where $n_P \in \mathbb{Z}$ and only finitely many n_P are non-zero. n_P is called the **multiplicity** of P in D.

Definition 1.3.0.2. The **degree** of a divisor D is given by $\deg(D) = \sum_{P \in C} n_P$, i.e., it is the summation of multiplicities of all points in C.

Example 1.3.0.3. A fundamental example of a divisor is the **principal divisor** associated to a non-zero rational function f on C. We can define the divisor of f as

$$\operatorname{div}(f) = \sum_{P \in C} \operatorname{ord}_P(f) P$$

where $\operatorname{ord}_P(f)$ is the order of vanishing of f at P. If $\operatorname{ord}_P(f) > 0$, P is a **zero** of f, and if $\operatorname{ord}_P(f) < 0$, P is a **pole** of f. A well-known fact about principal divisors on a smooth projective curve is that the number of zeroes equals the number of poles (counted with multiplicity), which implies $\deg(\operatorname{div}(f)) = 0$.

Example 1.3.0.4. Consider the complex projective line $C = \mathbb{P}^1$ and the rational function $f(z) = \frac{(z-1)^2}{z-2}$. This function has a zero of order 2 at $z = 1$, a pole of order 1 at $z = 2$, and a pole of order 1 at $z = \infty$. The principal divisor associated to f is

$$\operatorname{div}(f) = 2[1] - 1[2] - 1[\infty].$$

The degree of this divisor is clearly $\deg(\operatorname{div}(f)) = 2 - 1 - 1 = 0$.

Theorem 1.3.0.5 (Riemann-Hurwitz Formula)**.** *Let X and Y be smooth curves. Assume $\phi : X \to Y$ is a degree d map*[1]*. Then*

$$2g(X) - 2 = d(2g(Y) - 2) + \sum_{P \in X} (e_P - 1),$$

where e_P is the ramification index *of P and $g(X), g(Y)$ denotes the* genus *of X and Y, respectively.*

Example 1.3.0.6. Consider a hyperelliptic curve X of genus g and the projective line $Y = \mathbb{P}^1$. There exists a degree $d = 2$ covering map $\phi : X \to \mathbb{P}^1$. For this map, the genus of the target is $g(\mathbb{P}^1) = 0$. The map ϕ ramifies at precisely $2g + 2$ points (the Weierstrass points of X), and at each such point P, the ramification index is $e_P = 2$. Thus, $\sum_{P \in X}(e_P - 1) = (2g + 2)(2 - 1) = 2g + 2$.

Plugging these values into the Riemann-Hurwitz formula yields:

$$2g(X) - 2 = 2(2(0) - 2) + (2g + 2) = -4 + 2g + 2 = 2g - 2,$$

which verifies the formula.

[1] The degree d of a map ϕ is the number of preimages of a generic point in Y, or equivalently, the degree of the function field extension $[K(X) : \phi^* K(Y)]$.

Theorem 1.3.0.7 (Degree-genus formula)**.** *Let C be a smooth curve of degree d in $\mathbb{P}^n$. Then*

$$g(C) = \frac{(d-1)(d-2)}{2}$$

Theorem 1.3.0.8 (Riemann-Roch Theorem)**.** *Let C be a smooth projective curve of genus g and D be a divisor of C. Then*

$$h^0(C, D) - h^1(C, D) = \deg(D) - g + 1,$$

where $h^i(C, D) := \dim H^i(C, D)$ and $H^i(C, D)$ is the i-th cohomology group of the sheaf $\mathcal{O}_C(D)$.

Theorem 1.3.0.9 (Clifford's Theorem)**.** *If $l(D) > 0$, and $l(K_C - D) > 0$, then*

$$l(D) \leq \frac{deg(D)}{2} + 1$$

Theorem 1.3.0.10 (Serre duality)**.** *Let C be a smooth curve and D is a divisor of C, then*

$$H^1(D) = H^0(K_C - D)^\vee := \mathrm{Hom}_{\mathbb{C}}(H^0(K_C - D), \mathbb{C})$$

and in particular, $h^1(D) = h^0(K_C - D)$.

Theorem 1.3.0.11 (Serre - Grothendieck vanishing theorem)**.** *Let $\mathcal{F}$ is a coherent sheaf on a projective scheme X and $dim(X) = n$. Then, for any i, $H^i(\mathcal{F})$ is finite dimensional and it is 0 if $i > n$. Moreover, if $X \subset \mathbb{P}^k$ then for $d \gg 0$, $\mathcal{F}(d)$ is generated by its global sections and $H^i(\mathcal{F}(d)) = 0$ for $i > 0$.*

1.4 Hyperelliptic Curves

Definition 1.4.0.1. A **hyperelliptic curve** of genus $g \geq 2$ is a smooth projective curve admitting a degree 2 map $\phi : C \to \mathbb{P}^1$.

Example 1.4.0.2. The affine curve given by the equation $y^2 = x^5 - x$ can be compactified to yield a hyperelliptic curve C of genus $g = 2$. The degree 2 map $\phi : C \to \mathbb{P}^1$ is given on the affine patch by the projection $(x, y) \mapsto [x : 1]$ in homogeneous coordinates of $\mathbb{P}^1$. This map extends to the unique point at infinity on C, which is mapped to the point at infinity $[1 : 0] \in \mathbb{P}^1$.

Lemma 1.4.0.3. *Let C be a hyperelliptic curve of genus $g \geq 2$. Then C has exactly $2g+2$ ramification points.*

Proof. Let C be a hyperelliptic curve of genus $g \geq 2$. Then it has a 2-to-1 map $\phi : C \to \mathbb{P}^1$. Then by Riemann-Hurwitz formula, we have:

$$2g(C) - 2 = deg(\phi)(2g(\mathbb{P}^1) - 2) + \sum_{P \in C}(deg(\phi_P) - 1)$$

We know that $\mathbb{P}^1$ has genus 0. Thus, $g(\mathbb{P}^1) = 0$. Then

$$\begin{aligned} 2g(C) - 2 &= 2(0-2) + \sum_{P \in C}(2-1) \\ 2g - 2 &= -4 + \sum_{P \in C} 1 \\ 2g + 2 &= \sum_{P \in C} 1 \end{aligned}$$

Thus, there exists exactly $2g+2$ ramification points.

□

Definition 1.4.0.4. The **hyperelliptic involution** $\iota : C \to C$ is the unique automorphism of C of order 2 with exactly $2g+2$ fixed points. The quotient $C/\{1, \iota\}$ is $\mathbb{P}^1$.

Definition 1.4.0.5. Let C be a smooth projective curve of genus $g \geq 2$ and $\{w_1, \ldots, w_g\}$ be a basis for ω_C. Then the mapping

$$\phi_{\omega_C} : C \to \mathbb{P}^{g-1}, \quad P \mapsto [w_1(P) : \ldots : w_g(P)]$$

is called the **canonical map** of C.

Proposition 1.4.0.6. *If ϕ_{ω_C} is not injective, then C is hyperelliptic.*

Proof. Let $p, q \in C$ be two distinct points and let $\phi_{\omega_C}(p) = \phi_{\omega_C}(q)$. This means that

$$w_\alpha(p) = \lambda w_\alpha(q) \quad \forall \alpha = 1, \ldots, g, \quad \lambda \neq 0$$

Let $D = p + q$. Apply Riemann-Roch to the divisor D, then we get

$$\begin{aligned} l(D) - l(K_C - D) &= deg(D) + 1 - g \\ l(D) - (g-1) &= 2 + 1 - g \\ l(D) &= 2 + 1 - g + g - 1 = 2 \end{aligned}$$

Thus, there exists a nonconstant map $f : C \to \mathbb{P}^1$. We have $f^{-1}(\infty) = p + q$ because if it was a single point p (or q) we would have $C \cong \mathbb{P}^1$. Therefore, $\deg(f) = 2$ and C is hyperelliptic.

□

Lemma 1.4.0.7. *All genus 2 curves are hyperelliptic.*

Proof. Let C be a genus 2 curve and $\phi : C \to \mathbb{P}^1$. Then again by Riemann-Hurwitz formula, we have:

$$2g - 2 = \deg(\phi)(2g(\mathbb{P}^1) - 2) + \sum_{P \in C}(deg(\phi_P) - 1)$$

We know that $g = 2$ and $g(\mathbb{P}^1) = 0$. Thus,

$$4 - 2 = \deg(\phi)(0 - 2) + \sum_{P \in C}(deg(\phi_P) - 1)$$

By Proposition 1.4.0.6, we only need to show that the map ϕ is not injective.

$$\begin{aligned} 2 &= 1(0 - 2) + \sum_{P \in C}(1 - 1) \\ 2 &= -2 + 0 \\ 4 &\neq 0 \end{aligned}$$

Thus, the map ϕ cannot be injective. Therefore, C is hyperelliptic.

□

CHAPTER 2

PRELIMINARIES

2.1 Algebraic Groups

In this section, we will review the basic theory of algebraic groups. Main references for this section are [Mil22] and [Hos15].

For this section, we assume that every module over k is finitely generated. Also, we let $* = \mathrm{Spec}(k)$.

2.1.1 Basic Definitions

Definition 2.1.1.1. An **algebraic k-scheme** is a scheme of finite type over k.

Definition 2.1.1.2. Let G be an algebraic k-scheme, and let $m : G \times G \to G$ be a morphism of algebraic k-schemes. Then (G, m) is an **algebraic group** over k if there exists morphisms

$$e : * \to G, \qquad inv : G \to G$$

such that the following diagrams commute:

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{id_G \times m} & G \times G \\ {\scriptstyle m \times id_G}\downarrow & & \downarrow{\scriptstyle m} \\ G \times G & \xrightarrow[m]{} & G \end{array} \qquad \begin{array}{ccccc} * \times G & \xrightarrow{e \times id_G} & G \times G & \xleftarrow{id_G \times e} & G \times * \\ & \searrow^{\simeq} & \downarrow{\scriptstyle m} & \swarrow_{\simeq} & \\ & & G & & \end{array}$$

$$\begin{array}{ccccc} G & \xrightarrow{(inv, id_G)} & G \times G & \xleftarrow{(id_G, inv)} & G \\ \downarrow & & \downarrow{\scriptstyle m} & & \downarrow \\ * & \xrightarrow[e]{} & G & \xleftarrow[e]{} & * \end{array}$$

We say that G is a **group variety** (resp. **affine algebraic group**) if its underlying scheme G is a variety (resp. affine algebraic scheme).

Remark 2.1.1.3. In this thesis, we will mainly consider affine algebraic groups. If not specified otherwise, we will assume that algebraic groups are affine.

Example 2.1.1.4 (General Linear Group)**.** The general linear group GL_n is an affine algebraic group over k. Its underlying affine scheme is $\mathrm{Spec}(k[x_{11}, \ldots, x_{nn}, t]/(t\det(x_{ij}) - 1))$, where x_{ij} are the matrix coordinate functions and $\det(x_{ij})$ is the determinant polynomial. For any k-algebra R, its group of R-points $GL_n(R)$ is the group of invertible $n \times n$ matrices with entries in R.

Example 2.1.1.5 (Multiplicative Group)**.** The multiplicative group $\mathbb{G}_m$ is an affine algebraic group over k, which can be identified with GL_1. Its coordinate ring is $\mathcal{O}(\mathbb{G}_m) = k[x,t]/(xt-1) \cong k[x, x^{-1}]$. The comultiplication is given by $\Delta(x) = x \otimes x$, the counit by $e^*(x) = 1$, and the coinversion by $inv^*(x) = x^{-1}$. For any k-algebra R, $\mathbb{G}_m(R)$ is the group of units $R^\times$.

Example 2.1.1.6 (Additive Group)**.** The additive group $\mathbb{G}_a$ is an affine algebraic group over k whose underlying scheme is the affine line $\mathbb{A}^1_k = \mathrm{Spec}(k[x])$. The group structure is defined by the comultiplication $\Delta(x) = x \otimes 1 + 1 \otimes x$, the counit $e^*(x) = 0$, and the coinversion $inv^*(x) = -x$. For any k-algebra R, $\mathbb{G}_a(R)$ is simply the additive group of R.

Definition 2.1.1.7. A **homomorphism** of algebraic groups from G to H is a morphism $\phi : G \to H$ of algebraic schemes such that $\phi \circ m_G = m_H \circ (\phi \times \phi)$.

Definition 2.1.1.8. A **linear algebraic group** is an algebraic group which is a subgroup of GL_n for some $n \in \mathbb{N}$.

Definition 2.1.1.9. A **linear representation** of an algebraic group G on a vector space V over k is a morphism of group-valued functors (i.e., a functor from a category $\mathcal{C}$ to the category of groups $\mathbf{Grp}$) $\rho : G \to GL(V)$. If V is a finite dimensional vector space, this is equivalent to an algebraic group homomorphism $\rho : G \to GL(V)$, which is called a **finite dimensional linear representation** of G.

Definition 2.1.1.10. Let G be an affine algebraic group. Let $\mathcal{O}(G)$ be the coordinate ring of G. For an affine algebraic group G, the homomorphism of k-algebras Δ_G :

$\mathcal{O}(G) \to \mathcal{O}(G) \otimes_k \mathcal{O}(G)$ corresponding to the multiplication morphism $m : G \times G \to G$ is called the **comultiplication** of G. Similarly, the homomorphism of k-algebras $inv^* : \mathcal{O}(G) \to \mathcal{O}(G)$ corresponding to the inversion morphism $inv : G \to G$ is called the **coinversion** of G.

Theorem 2.1.1.11 (Cartier)**.** *Every affine algebraic group over a field of characteristic 0 is smooth.*

Proof. See [Mil22, Theorem 3.23]. □

2.1.2 Group Actions

Definition 2.1.2.1. Let G be an algebraic group and X be an algebraic scheme. An **action** of G on X is a morphism $\alpha : G \times X \to X$ such that the following diagrams commute:

$$\begin{array}{ccc} G \times G \times X & \xrightarrow{id_G \times \alpha} & G \times X \\ {\scriptstyle m_G \times id_X}\downarrow & & \downarrow{\scriptstyle \alpha} \\ G \times X & \xrightarrow[\alpha]{} & X \end{array} \qquad \begin{array}{ccc} * \times X & \xrightarrow{e \times id_X} & G \times X \\ & {\scriptstyle pr_2}\searrow & \downarrow{\scriptstyle \alpha} \\ & & X. \end{array}$$

Definition 2.1.2.2. Let α_X and α_Y be actions of an algebraic group G on algebraic schemes X and Y, respectively. A morphism $\phi : X \to Y$ is called G**-equivariant** if the following diagram commutes:

$$\begin{array}{ccc} G \times X & \xrightarrow{id_G \times \phi} & G \times Y \\ {\scriptstyle \alpha_X}\downarrow & & \downarrow{\scriptstyle \alpha_Y} \\ X & \xrightarrow[\phi]{} & Y. \end{array}$$

If Y is given the trivial action(i.e. $\alpha_Y(g, y) = y$ for all $g \in G, y \in Y$), then we say that ϕ is a G-invariant morphism.

Definition 2.1.2.3. If X is an algebraic scheme, G is an algebraic group with an action on X, and $\mathcal{O}(X)$ is the coordinate ring of X, then the action of G on X has an induced **coaction** homomorphism of k-algebras:

$$\begin{array}{rccc} \alpha^* : & \mathcal{O}(X) & \to & \mathcal{O}(G \times X) \cong \mathcal{O}(G) \otimes_k \mathcal{O}(X) \\ & f & \mapsto & \sum h_i \otimes f_i \end{array}$$

Definition 2.1.2.4. An action of an affine algebraic group G on a k-vector space V (resp. k-algebra A) is given by, for every k-algebra R, an action of $G(R)$ (R-points of G) on $V \otimes_k R$ (resp. $A \otimes_k R$)

$$\alpha_R : G(R) \times (V \otimes_k R) \to V \otimes_k R \qquad \text{resp. } \alpha_R : G(R) \times (A \otimes_k R) \to A \otimes_k R$$

such that $\alpha_R(g, -)$ is a morphism of R-modules (resp. R-algebras). Furthermore, this family of actions is *functorial* in R, meaning that for any k-algebra homomorphism $\phi : R \to S$, the following diagram commutes:

$$\begin{array}{ccc} G(R) \times (V \otimes_k R) & \xrightarrow{\alpha_R} & V \otimes_k R \\ \Big\downarrow{\scriptstyle G(\phi)\times(\mathrm{id}_V \otimes \phi)} & & \Big\downarrow{\scriptstyle \mathrm{id}_V \otimes \phi} \\ G(S) \times (V \otimes_k S) & \xrightarrow[\alpha_S]{} & V \otimes_k S \end{array}$$

Equivalently, for any $g \in G(R)$ and any $x \in V \otimes_k R$, we have the relation:

$$\alpha_S\big(G(\phi)(g),\, (\mathrm{id}_V \otimes \phi)(x)\big) = (\mathrm{id}_V \otimes \phi)\big(\alpha_R(g, x)\big). \tag{2.1}$$

An action of G on a k-algebra A is **rational** if every element of A is contained in some finite dimensional G-invariant linear subspace of A.

Lemma 2.1.2.5. *Assume G is an affine algebraic group acting on an affine algebraic scheme X. Then any $f \in \mathcal{O}(X)$ is contained in some finite dimensional G-invariant linear subspace of $\mathcal{O}(X)$. Furthermore, for any finite dimensional vector subspace W of $\mathcal{O}(G)$, there exists a finite dimensional G-invariant vector subspace V of $\mathcal{O}(X)$ containing W.*

Proof. Let $\alpha^* : \mathcal{O}(X) \to \mathcal{O}(G) \otimes_k \mathcal{O}(X)$ be the coaction of G on X. Let $f \in \mathcal{O}(X)$. Then $\alpha^*(f) = \sum h_i \otimes f_i$ for some $h_i \in \mathcal{O}(G)$ and $f_i \in \mathcal{O}(X)$. Then, $g \cdot f = \sum_i h_i(g) f_i$ and so the vector space spanned by $f_1, ..., f_n$ is a G-invariant vector subspace of $\mathcal{O}(X)$ containing f. The second part is obtained similarly by applying the same argument to a basis of W. □

Theorem 2.1.2.6. *Any affine algebraic group G over k is a linear algebraic group.*

Proof. Since G is an affine algebraic group, it is also an affine algebraic scheme over k. Thus, its coordinate ring $\mathcal{O}(G)$ is a finitely generated k-algebra. Therefore, the k-vector space W spanned by a choice of generators of $\mathcal{O}(G)$ is a k-vector space of finite dimension. By Lemma 2.1.2.5, there exists a finite dimensional G-invariant vector subspace V of $\mathcal{O}(G)$ containing W.

Let Δ_G be the comultiplication of G. Then, for a basis $f_1, \ldots, f_n$ of V, we have $\Delta_G(f_i) = \sum_{j=1}^{n} h_{ij} \otimes f_j$ for some $h_{ij} \in \mathcal{O}(G)$. For the action $\alpha : G \to \mathrm{Aut}(\mathcal{O}(G))$, we have $\alpha(g)(f_i) = \sum_j h_{ij}(g) f_j$. This gives a k-algebra homomorphism:

$$\rho^* : \mathcal{O}(\mathrm{Mat}_{n\times n}) \to \mathcal{O}(G), \qquad x_{ij} \mapsto h_{ij}.$$

We need to show that the induced morphism of affine schemes $\rho : G \to \mathrm{Mat}_{n\times n}$ is a closed embedding. Observe that V is contained in the image of ρ^*:

$$f_i = (\mathrm{Id}_{\mathcal{O}(G)} \otimes e^*)\Delta_G(f_i) = (\mathrm{Id}_{\mathcal{O}(G)} \otimes e^*) \sum_{j=1}^{n} h_{ij} \otimes f_j = \sum_{j=1}^{n} e^*(f_j) h_{ij}.$$

Observe that $e^*(f_j)$ are scalars in k. Therefore, the above equation expresses each basis element f_i as a k-linear combination of the elements h_{ij}. By definition, the image of a k-algebra homomorphism is a k-subalgebra. Thus, the image of ρ^* is a k-subalgebra of $\mathcal{O}(G)$, generated by the elements h_{ij}. Since this subalgebra is closed under k-linear combinations, it contains every f_i and consequently it contains the entire vector space V. Recall that V is chosen to contain a set of k-algebra generators of $\mathcal{O}(G)$. Since the image of ρ^* is a k-subalgebra that contains all of the generators of $\mathcal{O}(G)$, ρ^* must be surjective. Thus, ρ is a closed embedding.

Finally, we need to show that ρ is a semigroup homomorphism. To prove this, consider $m_G \circ (\mathrm{id}_G \times m_G) = m_G \circ (m_G \times \mathrm{id}_G)$ and apply to $f_i \in \mathcal{O}(G)$ to get

$$\sum_{k,j} h_{ik} \otimes h_{kj} \otimes f_j = \sum_{j} \Delta_G(h_{ij}) \otimes f_j.$$

This implies $\rho(g)\rho(h) = \rho(gh)$ for all $g, h \in G(k)$ and ρ is a semigroup homomorphism. Since G is a group, the image of ρ is a group, hence it is contained in $GL_n(k)$. Therefore, G is a linear algebraic group. □

2.1.3 Orbits and Stabilizers

Definition 2.1.3.1. Let G be an affine algebraic group acting on an algebraic scheme X by $\alpha : G \times X \to X$. Let $x \in X(k)$.

(i) The **orbit** $G \cdot x$ of x is the image of the $\alpha_x := \alpha(-, x) : G(k) \to X(k)$, $g \mapsto g \cdot x$.

(ii) The **stabiliser** G_x of x is the fibre product of $\alpha_x : G \to X$ and $x : * \to X$. In other words, we have the following commutative diagram:

$$\begin{array}{ccc} G_x & \longrightarrow & G \\ \downarrow & & \downarrow{\scriptstyle \alpha_x} \\ * & \xrightarrow[x]{} & X \end{array}$$

2.2 Stacks

In this section, we will review the basic theory of stacks, especially algebraic stacks and Deligne-Mumford stacks. Main references for this section are [Alp26] and [Ols16].

2.2.1 Sites and Sheaves on Sites

Definition 2.2.1.1. Let C be a category. A **Grothendieck topology** on C is a set $\mathrm{Cov}(X)$ of collections of morphisms $\{X_i \to X\}_{i \in I}$ for every $X \in C$ such that the following conditions are satisfied:

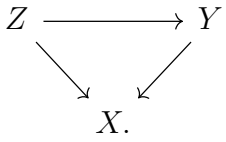
(i) If $Y \to X$ is an isomorphism, then $\{Y \to X\} \in \mathrm{Cov}(X)$.

(ii) If $\{X_i \to X\}_{i \in I} \in \mathrm{Cov}(X)$ and $Y \to X$ is a morphism, then the fiber products $X_i \times_X Y$ exist and the collection $\{X_i \times_X Y \to Y\}_{i \in I} \in \mathrm{Cov}(Y)$.

(iii) If $\{X_i \to X\}_{i \in I} \in \mathrm{Cov}(X)$ and $\{X_{ij} \to X_i\}_{j \in J_i} \in \mathrm{Cov}(X_i)$ for all $i \in I$, then the collection $\{X_{ij} \to X\}_{i \in I, j \in J_i} \in \mathrm{Cov}(X)$.

Definition 2.2.1.2. Let C be a category with a Grothendieck topology. Then, we say that C is a **site**.

Example 2.2.1.3 (Localized Category and Localized Site)**.** Let C be a category, $X \in C$. We define C/X as the **localized category** oof the objects over X whose objects are morphisms $Y \to X$ in C and whose morphisms are commutative diagrams



Furthermore, if C is a site,we may define $\mathrm{Cov}(Y \to X)$ as the collection of X-morphisms $\{Y_i \to Y\}_{i\in I}$ such that the underlying collection of morphisms $\{Y_i \to Y\}_{i\in I}$ in C, obtained by forgetting the morphisms to X, is in $\mathrm{Cov}(Y)$. Then, C/X is called the **localized site** of C at X.

Definition 2.2.1.4. A **presheaf** on a category C is a functor

$$F : C^{op} \to \mathbf{Set}$$

Definition 2.2.1.5. A **sheaf** on a site C is a presheaf F on C such that for every $U \in C$ and covering $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(U)$ the following sequence is exact:

$$F(U) \to \prod_{i\in I} F(U_i) \rightrightarrows \prod_{i,j\in I} F(U_i \times_U U_j)$$

In other words, $F(U)$ is the limit (or equalizer) of the following diagram:

$$\prod_{i\in I} F(U_i) \rightrightarrows \prod_{i,j\in I} F(U_i \times_U U_j)$$

Remark 2.2.1.6. If we change the target of F to **Grp** (resp. **Ab**, **Ring**, **Mod**), we get the definition of sheaves of groups (resp. sheaves of abelian groups, sheaves of rings, sheaves of modules) on a site C.

We have also another classical sheaf definition.

Definition 2.2.1.7. Let C be a site. A **sheaf** F on C is a presheaf on C satisfying the following two axioms for every $U \in C$ and every covering $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(U)$:

(i) **(Identity)** For any sections $s, t \in F(U)$, if their restrictions to each U_i agree (i.e., $s|_{U_i} = t|_{U_i}$ for all $i \in I$), then $s = t$.

(ii) **(Gluability)** Given sections $s_i \in F(U_i)$ for each $i \in I$ such that for any pair $i, j \in I$, their restrictions to $U_i \times_U U_j$ agree (i.e., $s_i|_{U_i \times_U U_j} = s_j|_{U_i \times_U U_j}$), then there exists a section $s \in F(U)$ such that $s|_{U_i} = s_i$ for all $i \in I$.

Here, the notation $s|_{U_i}$ refers to the image of s under the restriction map $F(U) \to F(U_i)$ induced by the morphism $U_i \to U$.

Remark 2.2.1.8. It can be easily seen that the two definitions are equivalent. The identity axiom is equivalent to exactness at the first position of the equalizer diagram. On the other hand, the gluability axiom is equivalent to exactness at the second position.

Example 2.2.1.9 (Category of Open Sets $\mathrm{Op}(X)$)**.** Let X be a topological space. The category $\mathrm{Op}(X)$ has open subsets of X as objects and inclusion maps as morphisms. We can define a Grothendieck topology on $\mathrm{Op}(X)$ by declaring that a family of morphisms $\{U_i \to U\}_{i \in I}$ is a covering if and only if $U = \bigcup_{i \in I} U_i$. With this topology, $\mathrm{Op}(X)$ becomes a site, and a sheaf on this site corresponds exactly to the standard definition of a sheaf on a topological space.

Example 2.2.1.10 (Sheaf of Differentiable Functions)**.** Let M be a differentiable manifold. We define a sheaf of rings $\mathcal{C}^\infty_M$ on the site $\mathrm{Op}(M)$ by assigning to each open set $U \subseteq M$ the ring of differentiable functions $f : U \to \mathbb{R}$. For an inclusion $U \subseteq V$, the restriction morphism $\mathcal{C}^\infty_M(V) \to \mathcal{C}^\infty_M(U)$ is given by the standard restriction of functions. Since differentiability is a local property, the identity and gluability axioms are naturally satisfied, making $\mathcal{C}^\infty_M$ a sheaf on $\mathrm{Op}(M)$.

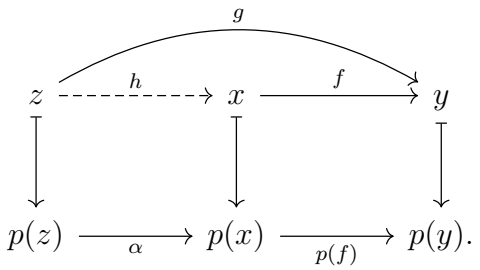
2.2.2 Fibered Categories and Categories Fibered in Groupoids

Definition 2.2.2.1. Let C be a category. A **category over** C is a category E together with a functor $p : E \to C$.
A morphism $f : x \to y$ in E is called **Cartesian** if for any morphism $g : z \to y$ in E and factorization

$$p(z) \xrightarrow{\alpha} p(x) \xrightarrow{p(f)} p(y)$$

of $p(g)$, there exists a unique $h : z \to x$ in E such that $p(h) = \alpha$ and $f \circ h = g$.
In other words, for the given solid arrows in the figure below, we get a dotted arrow

h making the diagram commute:



Remark 2.2.2.2. For a category (E, p) over C and object $X \in C$, we write $E(X)$ for the category with objects $x \in E$ such that $p(x) = X$ and morphisms $f : x \to y$ in E such that $p(f) = \mathrm{id}_X$.

Definition 2.2.2.3. A **fibered category** over a category C is a category E together with a functor $p_E : E \to C$ such that for every $\alpha : X' \to X$ in C and $x \in E(X)$, there exists a Cartesian morphism $f : x' \to x$ in E with $p_E(f) = \alpha$.

Definition 2.2.2.4. Let $p_E : E \to C$ and $p_F : F \to C$ be fibered categories over C. A **morphism of fibered categories** from E to F is a functor $\Phi : E \to F$ such that the following hold:

(i) $p_F \circ \Phi = p_E$ (equality of functors, not just an isomorphism).

(ii) Φ sends Cartesian morphisms in E to Cartesian morphisms in F.

Definition 2.2.2.5. If $\Phi, \Psi : E \to F$ are morphisms of fibered categories over C, then a **base preserving natural transformation** $\eta : \Phi \to \Psi$ is a natural transformation of functors such that for every $x \in E$ the morphism $\eta_x : \Phi(x) \to \Psi(x)$ in F projects to the identity morphism in C. We denote by

$$\mathrm{HOM}_C(E, F)$$

the category whose objects are morphisms of fibered categories from E to F and whose morphisms are base preserving natural transformations between them.

Proposition 2.2.2.6 (2-Yoneda Lemma)**.** *Let (E, p_E) be a fibered category over a category C. Let $X \in C$ and let C/X be the localized category of the objects over X. The localized category C/X is a fibered category over C with the forgetful functor $C/X \to C$ given by $(Y \to X) \mapsto Y$. The functor*

$$\xi : \mathrm{HOM}_C(C/X, E) \to E(X)$$

sending a morphism $g : C/X \to E$ to $g(id_X)$, is an equivalence of categories.

Proof. We begin by defining a functor

$$\eta : E(X) \to \mathrm{HOM}_C(C/X, E)$$

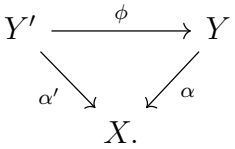
as follows: Let $x \in E(X)$, and for every morphism $\alpha : Y \to X$ in C choose a pullback $\alpha^* x \in E(Y)$ of x. Define

$$\eta_x : C/X \to E$$

by

$$(\alpha : Y \to X) \mapsto \alpha^* x.$$

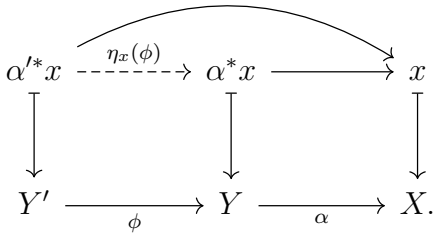
Given a morphism



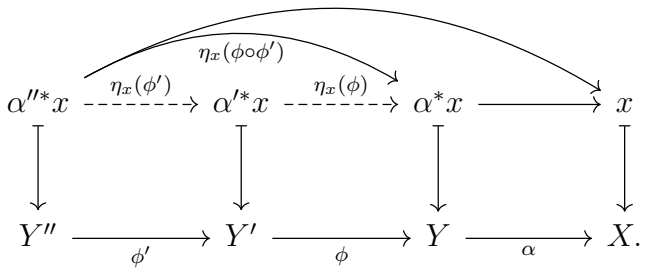
in C/X, define

$$\eta_x(\phi) : \alpha'^* x \to \alpha^* x$$

to be the unique morphism making the following diagram commute:



First, we need to verify that $\eta_x(\phi)$ is compatible with composition. Consider a sequence of morphisms: $Y'' \xrightarrow{\phi'} Y' \xrightarrow{\phi} Y$ in C/X. We need to show that $\eta_x(\phi \circ \phi') = \eta_x(\phi) \circ \eta_x(\phi')$. Consider the following pasted commutative diagram mapping the entire composition:



Both dotted morphisms exist separately by the universal property of Cartesian morphisms. Since each individual rectangle is commutative, the entire diagram is commutative. By the uniqueness of the Cartesian lifts, $\eta_x(\phi \circ \phi') = \eta_x(\phi) \circ \eta_x(\phi')$. Thus, η_x is compatible with composition. Given a morphism $f : x' \to x$ in $E(X)$, we have a morphism of functors $\eta_f : \eta_{x'} \to \eta_x$ defined by $\eta_f(\alpha : Y \to X) = \alpha^* f$.
For any morphism $\alpha : Y \to X$ in C/X and pullbacks $\alpha^* x$ and $\alpha^* x'$ of x and x', we define $\eta_f(\alpha)$ as the unique morphism filling the following diagram:

$$\begin{array}{ccc} \alpha^* x' & \xrightarrow{\eta_f(\alpha)} & \alpha^* x \\ \downarrow & & \downarrow \\ x' & \xrightarrow{f} & x. \end{array}$$

This way, we get a morphism of functors $\eta_f : \eta_{x'} \to \eta_x$, and in the end we get a functor $\eta : E(X) \to \mathrm{HOM}_C(C/X, E)$, as we first defined. The composite functor

$$E(X) \xrightarrow{\eta} \mathrm{HOM}_C(C/X, E) \xrightarrow{\xi} E(X)$$

takes $x \in E(X)$ to $\xi(\eta(x)) \cong \xi(\eta_x) \cong \eta_x(id_X) \cong id_X^* x \cong x$. This defines an isomorphism of functors

$$\xi \circ \eta \cong \mathrm{id}_{E(X)}.$$

Next, we consider the following composition of functors

$$\mathrm{HOM}_C(C/X, E) \xrightarrow{\xi} E(X) \xrightarrow{\eta} \mathrm{HOM}_C(C/X, E).$$

This composition takes $g : C/X \to E$ to $\eta(\xi(g)) \cong \eta(g(id_X)) \cong g$. This defines an isomorphism of functors

$$\eta \circ \xi \cong \mathrm{id}_{\mathrm{HOM}_C(C/X,E)}.$$

□

Corollary 2.2.2.7. *Let $X, Y \in C$. Then the functor*

$$\mathrm{HOM}_C((C/X), (C/Y)) \to \mathrm{Hom}_C(X, Y), \qquad f \mapsto f(id_X)$$

is an equivalence of categories.

Proof. Take $E = (C/Y)$ in the 2-Yoneda Lemma. □

Remark 2.2.2.8. The 2-Yoneda Lemma plays a fundamental role in moduli theory. If E is a fibered category representing a moduli problem (where $E(X)$ is the groupoid of families over a base scheme X), the lemma states that there is a natural equivalence between giving a family over X and giving a morphism from X (viewed as the localized category C/X) to E. In other words, assigning a family of geometric objects (like curves or vector bundles) over X is equivalent to giving a classifying morphism $X \to E$. Furthermore, if E is representable by a fine moduli space M, the identity morphism $\mathrm{id}_M \in \mathrm{Hom}(M, M)$ corresponds to the *universal family* over M, and every other family is uniquely obtained as a pullback of this universal family.

Definition 2.2.2.9. A **category fibered in groupoids** over a category C is a fibered category (E, p_E) such that for every $X \in C$ the category $E(X)$ is a groupoid.

2.2.3 Stacks and Algebraic Spaces

Definition 2.2.3.1. A category fibered in groupoids (E, p_E) over a category C is called a **stack** if for every $X \in C$ and covering $\{X_i \to X\}_{i \in I}$, the functor

$$E(X) \to E(\{X_i \to X\}_{i \in I})$$

is an equivalence of categories.

There are several equivalent definitions of stacks.The most common one is from [DM69, 4.1]:

Definition 2.2.3.2. Let C be a site. A **stack** over C is a category fibered in groupoids $p : E \to C$ satisfying the following conditions:

(i) For any object $X \in C$ and any objects $x, y \in E(X)$ the functor from localized category C/X to (sets) which to any morphism $\phi : Y \to X$ associates $\mathrm{Hom}_{E(Y)}(\phi^* x, \phi^* y)$ is a sheaf.

(ii) For any covering $\{X_i \to X\}$ of an object $X \in C$, any descent data with respect to this covering is effective.

Definition 2.2.3.3. A category fibered in groupoids $p : E \to C$ is called a **prestack** if it satisfies the condition (i) of Definition 2.2.3.2.

Theorem 2.2.3.4 (Stackification)**.** *Let E be a prestack over a site C. Then there exists a stack E^{st}, called **stackification**, and a prestack morphism $\phi : E \to E^{st}$ such that*

1. *for every $U \in ob(C)$, and any $x, y \in E(U)$, the functor*

$$Hom(x, y) \to Hom(\phi(x), \phi(y))$$

   induced by ϕ identifies the right hand side with the sheafification of the left hand side.

2. *for every $U \in ob(C)$, and any $x' \in ob(E^{st}(U))$ there exists a covering $\{U_i \to U\}_{i \in I}$ such that for every $i \in I$ the object $x'|_{U_i}$ is in the essential image of the functor $\phi : E(U_i) \to E^{st}(U_i)$.*

Proof. See [Sta26, Lemma 8.9.1]. □

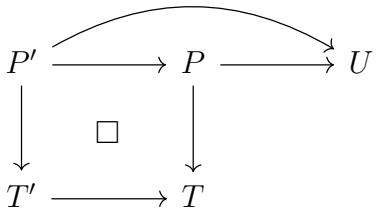
Definition 2.2.3.5 (Classifying stacks)**.** The **classifying stack** BG of a smooth affine algebraic group $G \to S$ is the category over Sch_S (S-schemes) whose objects are principal G-bundles $P \to X$ and a morphism $(P' \to T') \to (P \to T)$ is the data of a G-equivariant morphism $P' \to P$ such that the following diagram is cartesian:

$$\begin{array}{ccc} P' & \longrightarrow & P \\ \downarrow & \square & \downarrow \\ T' & \longrightarrow & T \end{array}$$

Definition 2.2.3.6 (Quotient prestacks and stacks)**.** Let $G \to S$ be a smooth affine algebraic group acting on an S-scheme U. The **quotient prestack** $[U/G]^{pre}$ is the category over Sch_S consisting of pairs (T, u) where T is an S-scheme and $u \in U(T)$. A morphism $(T', u') \to (T, u)$ is the data of a morphism $f : T' \to T$ and an element $g \in G(T')$ such that $f^*u = g \cdot u'$. The **quotient stack** $[U/G]$ is the stackification of $[U/G]^{pre}$. In other words, $[U/G]$ is the category over Sch_S consisting of diagrams of the form:

$$\begin{array}{ccc} P & \longrightarrow & U \\ \downarrow & & \\ T & & \end{array}$$

where $P \to T$ is a principal G-bundle and $P \to U$ is a G-equivariant morphism of S-schemes. A morphism $(T' \leftarrow P' \to U) \to (T \leftarrow P \to U)$ consists of morphism $T' \to T$ and a G-equivariant morphism $P' \to P$ such that



is commutative and the left square is cartesian.

In some parts after this point, we will write T instead of its functor of points h_T by abuse of notation.

Definition 2.2.3.7. A morphism $X \to Y$ of prestacks (or presheaves) over Sch is **schematic** (or **representable by schemes**) if $X \times_Y T$ is a scheme for all morphism $T \to Y$ where T is a scheme.

Definition 2.2.3.8. If $\mathcal{P}$ is a property of morphisms of schemes stable under base change, a morphism $X \to Y$ of schematic prestacks has the same property if the morphism $X \times_Y T \to T$ has property $\mathcal{P}$ for all morphisms $T \to Y$ where T is a scheme.

Definition 2.2.3.9 (Algebraic Spaces)**.** An **algebraic space** is a sheaf X on $Sch_{\text{ét}}$ such that there exists a scheme U and a schematic surjective étale morphism $U \to X$.

Remark 2.2.3.10. The morphism $U \to X$ is called an **étale presentation**. Morphisms of algebraic spaces are morphism of sheaves. Every scheme is an algebraic space.

Definition 2.2.3.11. A morphism $X \to Y$ of prestacks (or presheaves) over Sch is called **representable** if the fiber product $X \times_Y T$ is an algebraic space for every morphism $T \to Y$ where T is a scheme.

Remark 2.2.3.12. Every schematic morphism is representable, since every scheme is an algebraic space.

Definition 2.2.3.13 (Deligne-Mumford Stacks)**.** A **Deligne-Mumford stack** is a stack X over $Sch_{\text{ét}}$ such that there exists a scheme U and a representable, étale, surjective morphism $U \to X$.

Remark 2.2.3.14. The morphism $U \to X$ is called an **étale presentation**. Morphisms of Deligne-Mumford stacks are morphisms of stacks. Every scheme or algebraic space is a Deligne-Mumford stack.

Definition 2.2.3.15 (Algebraic Stacks)**.** An **algebraic stack** (or **Artin stack**) is a stack X over $Sch_{\text{ét}}$ such that there exists a scheme U and a surjective, representable, and smooth morphism $U \to X$.

Remark 2.2.3.16. The morphism $U \to X$ is called a **smooth presentation**. Morphisms of algebraic stacks are morphisms of prestacks. Every scheme, algebraic space, or Deligne-Mumford stack is an algebraic stack.

Theorem 2.2.3.17. *Let X be an algebraic stack over a scheme S. Then X is a Deligne-Mumford stack if and only if the diagonal*

$$\delta : X \to X \times_S X$$

is formally unramified (i.e. the objects of X admit no infinitesimal automorphisms)

Proof. See [Ols16, Theorem 8.3.3] □

Theorem 2.2.3.18. *If $G \to S$ is a smooth affine group scheme acting on an algebraic space $U \to S$, the quotient stack $[U/G]$ is an algebraic stack over S such that $U \to [U/G]$ is a principal G-bundle and in particular surjective, smooth, and affine. Also, the classifying stack BG is an algebraic stack.*

Proof. Let $U \to [U/G]$ be the canonical projection. By the 2-Yoneda lemma, it corresponds to the trivial principal G-bundle $p_2 : G \times U \to U$ and the G-equivariant morphism $\mu : G \times U \to U$ which is given by multiplication. Let $T \to [U/G]$ be a morphism from an S-scheme classified by a principal G-bundle $P \to T$ and a G-equivariant morphism $f : P \to U$. We need to show that the following diagram is cartesian:

$$\begin{array}{ccc} P & \longrightarrow & U \\ \downarrow & \square & \downarrow \\ T & \longrightarrow & [U/G] \end{array}$$

To prove that the diagram is 2-cartesian, let X be an S-scheme. The groupoid $[U/G](X)$ consists of pairs $(Y \to X, \phi : Y \to U)$ where Y is a principal G-bundle and ϕ is a G-equivariant morphism. By the definition of 2-fiber products, a morphism $X \to T \times_{[U/G]} U$ is given by a triple $(t : X \to T, u : X \to U, \alpha)$, where $\alpha : t^*P \xrightarrow{\sim} X \times_S G$ is an isomorphism of G-bundles over X.

Let us explicitly write down the G-equivariant maps to U corresponding to the pullbacks of $T \to [U/G]$ and $U \to [U/G]$ to X:

1. Pullback via $t : X \to T$: The bundle is $t^*P \to X$ and the equivariant map is $f \circ \bar{t} : t^*P \to U$, where $\bar{t} : t^*P \to P$ is the canonical projection.
2. Pullback via $u : X \to U$: The bundle is the trivial bundle $X \times_S G \to X$ and the equivariant map is $\Phi_u : X \times_S G \to U$, given by $\Phi_u(x, g) = g \cdot u(x)$.

The 2-isomorphism α must be compatible with these equivariant maps, meaning $\Phi_u \circ \alpha = f \circ \bar{t}$.
Now, let $s : X \to t^*P$ be the global section defined by $s(x) = \alpha^{-1}(x, e)$. By the universal property of the pullback square, giving a section of $t^*P \to X$ is equivalent to giving a morphism $p : X \to P$ such that the composition $X \xrightarrow{p} P \to T$ is exactly t. This correspondence is simply given by $p = \bar{t} \circ s$.

Applying the compatibility condition to the section s, we compute:

$$\begin{aligned} u(x) &= e \cdot u(x) \\ &= \Phi_u(x, e) \\ &= \Phi_u(\alpha(s(x))) \\ &= (\Phi_u \circ \alpha)(s(x)) \\ &= (f \circ \bar{t})(s(x)) \\ &= f(\bar{t}(s(x))) \\ &= f(p(x)) \\ &= (f \circ p)(x) \end{aligned}$$

This demonstrates that $u = f \circ p$, meaning u is entirely determined by the morphism p. Thus, we obtain a bijection $(T \times_{[U/G]} U)(X) \cong P(X)$ given by sending the triple

(t, u, α) to p. By the Yoneda Lemma, we conclude that $T \times_{[U/G]} U \cong P$. Thus, the diagram is cartesian.

Since the pullback of $U \to [U/G]$ along any morphism $T \to [U/G]$ from a scheme is a principal G-bundle $P \to T$, the morphism $U \to [U/G]$ is representable, surjective, and smooth. Since U is an algebraic space it has an étale presentation $U' \to U$. Therefore, the composition $U' \to U \to [U/G]$ is a smooth presentation from a scheme, proving that $[U/G]$ is an algebraic stack. Setting $U = S$ with the trivial G-action, we deduce that the classifying stack $BG = [S/G]$ is also an algebraic stack. $\square$

CHAPTER 3

GEOMETRIC INVARIANT THEORY

Geometric Invariant Theory (GIT), originally developed by David Mumford in the 1960s, is a mathematical framework used to construct well-behaved quotients in algebraic geometry. In modern moduli theory, a common approach to parameterizing geometric objects is to first construct a rigidified parameter space (such as a Hilbert or Chow scheme) and then take the quotient by the action of an algebraic group that accounts for isomorphisms between these objects. However, taking the naive orbit space often results in a space with pathologically non-separated topologies due to the presence of non-closed orbits. GIT provides a systematic method to overcome these topological hurdles by identifying a locus of "well-behaved" points—known as stable and semistable points—whose quotient naturally admits the structure of a separated algebraic scheme.

It is important to emphasize that GIT is not a universal mechanism that perfectly parametrizes every single object in a given moduli problem. Because it requires discarding "unstable" points to achieve a well-behaved algebraic structure, the resulting GIT quotient only classifies the semistable objects, meaning some degenerate or highly singular configurations are inherently excluded from the final moduli space. Furthermore, the quotienting process often forces distinct but closely related semistable objects (whose orbits share points in their closures) to be identified as the same point in the quotient. Consequently, a GIT quotient should be understood not as a perfect set-theoretic orbit space of all possible objects, but rather as the best possible projective algebraic approximation for the moduli space.

3.1 Hilbert Schemes

In this section, we will review the basic theory of Hilbert schemes. The main references for this section are [FGI+05] and [ACG11].

Definition 3.1.0.1. The **Hilbert functor** $h_P : Sch^{op} \to Sets$ is the functor of flat families of schemes in $\mathbb{P}^n_{\mathbb{Z}}$ with Hilbert polynomial P, which associates to any scheme S the set of subschemes $X \subset \mathbb{P}^n_S$ whose fibers over the points of S have Hilbert polynomial P.

Theorem 3.1.0.2. *The **Hilbert scheme** $\mathrm{Hilb}_n^{P(m)}$ is a projective scheme that represents the Hilbert functor, i.e.,* $\mathrm{Hom}_{Sch}(\text{-}, \mathrm{Hilb}_n^{P(m)}) \cong h_P$.

In this thesis, we will not concern ourselves with the construction, existence, and projectivity of Hilbert schemes. For more details, we refer the reader to the foundational paper by Grothendieck [Gro61].

Definition 3.1.0.3. Let X be a projective scheme over a field k (with a chosen very ample line bundle $\mathcal{O}_X(1)$) and $\mathcal{F}$ a coherent sheaf on X. Then its **Hilbert polynomial** is:

$$P_{\mathcal{F}}(m) = \chi(X, \mathcal{F}(m)) = \sum_{i=0}^{\dim X} (-1)^i h^i(X, \mathcal{F}(m))$$

Example 3.1.0.4 (Hilbert Polynomial of $\mathbb{P}^n$)**.** Let $X = \mathbb{P}^n_k$ with the standard twisting sheaf $\mathcal{O}_{\mathbb{P}^n}(1)$, and let $\mathcal{F} = \mathcal{O}_{\mathbb{P}^n}$. For any $m \geq 0$, by Serre's calculation of the cohomology of projective space, $H^i(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m)) = 0$ for $i > 0$. The space of global sections $H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m))$ corresponds to homogeneous polynomials of degree m in $n+1$ variables. Thus, the Hilbert polynomial is given by:

$$P_{\mathcal{O}_{\mathbb{P}^n}}(m) = \binom{m+n}{n}.$$

Example 3.1.0.5 (Hilbert Polynomial of a Hypersurface)**.** Let $X \subset \mathbb{P}^n_k$ be a hypersurface of degree d, and let $\mathcal{F} = \mathcal{O}_X$. We have the standard short exact sequence of sheaves on $\mathbb{P}^n$:

$$0 \to \mathcal{O}_{\mathbb{P}^n}(-d) \to \mathcal{O}_{\mathbb{P}^n} \to \mathcal{O}_X \to 0.$$

Twisting by $\mathcal{O}_{\mathbb{P}^n}(m)$ preserves exactness. Using the additivity of the Euler characteristic on exact sequences, the Hilbert polynomial of X is the difference of the Hilbert

polynomials of $\mathbb{P}^n$ and $\mathbb{P}^n$ shifted by $-d$:

$$P_{\mathcal{O}_X}(m) = \chi(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m)) - \chi(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(m-d)) = \binom{m+n}{n} - \binom{m-d+n}{n}.$$

Proposition 3.1.0.6. *Let C be a hyperelliptic curve. Since the dualizing sheaf of C is not very ample, we need to use a ν-canonical bundle of C, which is a ν-th tensor power of the dualizing sheaf ω_C of C, for some integer $\nu \geq 3$. Thus, the Hilbert polynomial of C is given by $P(m) = (2m\nu - 1)(g-1)$.*

Proof. Let C be a hyperelliptic curve of genus g and ω_C be its dualizing sheaf. By the Riemann-Roch theorem for curves, we have:

$$h^0(C, \omega_C^{\otimes m\nu}) - h^0(C, K_C \otimes \omega_C^{-\otimes m\nu}) = \deg(\omega_C^{\otimes m\nu}) + (1-g) \tag{3.1}$$

By Serre duality, $h^0(C, K_C \otimes \omega_C^{-\otimes m\nu}) = h^1(C, \omega_C^{\otimes m\nu})$. The left-hand side of (3.1) is equal to the Euler characteristic $P(m) = \chi(C, \omega_C^{\otimes m\nu})$. Note that $h^i(C, \omega_C^{\otimes m\nu}) = 0$ for $i \geq 2$ by Serre-Grothendieck vanishing theorem. Evaluating the right-hand side of (3.1), we get:

$$\begin{aligned} P(m) &= \deg(\omega_C^{\otimes m\nu}) + (1-g) \\ &= m\nu \deg(\omega_C) + (1-g) \\ &= m\nu(2g-2) + (1-g) \\ &= 2m\nu(g-1) - (g-1) \\ &= (2m\nu - 1)(g-1) \end{aligned}$$

□

Remark 3.1.0.7. We need to find the dimension N of the projective space $\mathbb{P}^N$ into which our curve C is embedded by $\omega_C^{\otimes\nu}$. We have $N = h^0(C, \omega_C^{\otimes\nu}) - 1 = (2\nu - 1)(g-1) - 1$, since $h^1(C, \omega_C^{\otimes\nu}) = 0$ by Serre duality and since $deg(K_C \otimes \omega_C^{-\otimes\nu}) < 0$.

3.2 Basics of Geometric Invariant Theory

3.2.1 Good Quotients

When an algebraic group G acts on a scheme S, the naive set-theoretic orbit space S/G often fails to admit the structure of an algebraic scheme, largely because non-

closed orbits can lead to a non-separated topology. To resolve this, Geometric Invariant Theory introduces the notion of a *good quotient*. This provides a well-behaved scheme structure whose coordinate ring consists of G-invariant functions, effectively identifying orbits whose closures intersect to ensure a separated algebraic space.

Definition 3.2.1.1. Let G be a linearly reductive group acting on an algebraic space S over $\mathbb{C}$. Then, a G-invariant map $\pi : S \to X$ is called a **good quotient** if:

1. $\mathcal{O}_X \to (\pi_* \mathcal{O}_S)^G$ is an isomorphism
2. π is affine

The good quotient of S by G is denoted by $S//G = X$.

Example 3.2.1.2. Consider the natural action of the multiplicative group $G = \mathbb{C}^*$ on the affine space $S = \mathbb{A}^n_{\mathbb{C}}$ by scalar multiplication, where $\lambda \cdot (x_1, \ldots, x_n) = (\lambda x_1, \ldots, \lambda x_n)$. The coordinate ring of S is $\mathbb{C}[x_1, \ldots, x_n]$, and the action of G grades this ring by polynomial degree. The ring of invariants $\mathbb{C}[x_1, \ldots, x_n]^G$ consists solely of the degree-zero polynomials, which are just the constants $\mathbb{C}$. Thus, the good quotient is $S//G = \mathrm{Spec}(\mathbb{C})$, a single point. Geometrically, the orbits of this action are the origin $\{0\}$ and the punctured lines through the origin. Since the closure of every punctured line contains the origin, all orbits share a point in their closure. The good quotient identifies them all into a single point, illustrating that good quotients separate closed orbits rather than all orbits.

While a good quotient guarantees a well-behaved algebraic structure, it may collapse multiple distinct orbits into a single point, thereby losing geometric information about the original action. To remedy this, we introduce the *geometric quotient*, which strengthens the good quotient by ensuring that it strictly parametrizes the actual orbits. This provides a true orbit space in the category of algebraic schemes, generalizing the intuitive notion of a set-theoretic quotient.

Definition 3.2.1.3. The **geometric quotient** of S by G is a morphism $f : S \to X$ of algebraic schemes, satisfying the following conditions:

1. f is a good quotient of S by G.

2. For every $x \in X$, the fiber $f^{-1}(x)$ consists of a single orbit.

Remark 3.2.1.4. The orbits in the previous definition are closed.

Definition 3.2.1.5. Let G be a linearly reductive group acting on a projective $\mathbb{C}$-scheme S and let $\mathcal{O}_S(1)$ be an ample line bundle with a G-action. Then the **stable locus** and the **semistable locus** are defined as follows:

$$S^{ss} := \{s \in S \mid \exists f \in \Gamma(S, \mathcal{O}_S(d))^G \text{ with } d > 0 \text{ such that } f(s) \neq 0\}$$

$$S^s := \left\{ s \in S \;\middle|\; \begin{array}{l} \text{there exists } f \in \Gamma(S, \mathcal{O}_S(d))^G \text{ with } d > 0 \text{ such that} \\ \quad \bullet\ f(s) \neq 0, \\ \quad \bullet\ \text{the orbit } G \cdot s \subseteq S_f \text{ is closed, and} \\ \quad \bullet\ \text{the function } S \to \mathbb{Z},\ x \mapsto \dim G_x \text{ is constant in an open} \\ \qquad \text{neighborhood of } s \end{array} \right\}.$$

A point $s \in S$ is called **stable** if $s \in S^s$ and **semistable** if $s \in S^{ss}$.

Theorem 3.2.1.6. *Let G be a linearly reductive group over $\mathbb{C}$. Let $U \subseteq \mathbb{P}(V)$ be a G-equivariant closed subscheme, where V is a finite-dimensional representation of G. Then the following diagram is cartesian:*

$$\begin{array}{ccccc} U^s & \hookrightarrow & U^{ss} & \hookrightarrow & U \\ \downarrow & \square & \downarrow{\scriptstyle \widetilde{\pi}} & & \\ U^s/G & \hookrightarrow & U^{ss}//G & & \end{array}$$

where $U^s/G \subseteq U^{ss}//G$ is an open subscheme and the map $\widetilde{\pi}$ is the good quotient of U^{ss} by G and the restriction to U^s is the geometric quotient. If also, the action of G on U has finite stabilizers, $[U^s/G]$ is separated.

Proof. See [Alp26, Theorem 7.2.6] □

Definition 3.2.1.7. A **1-parameter subgroup** of G is a homomorphism $\lambda : \mathbb{G}_m \to G$.

Definition 3.2.1.8. Let $U \subseteq \mathbb{P}(V)$ be a G-equivariant closed subscheme, where V is a finite-dimensional representation of G and $u \in U(\mathbb{C})$ with a lift $\tilde{u} \in \mathbb{A}(V)$. Given a 1-PS λ, we choose a basis for $V \simeq \mathbb{C}^n$ such that $\lambda(t) \cdot (v_1, ..., v_n) = (t^{a_1} v_1, ..., t^{a_n} v_n)$ for some integers a_i. The **Hilbert-Mumford index** of u with respect to λ is defined as

$$\mu(u, \lambda) = \max_{i, \tilde{u}_i \neq 0} -a_i$$

Remark 3.2.1.9. By the definition of the Hilbert-Mumford index, we get:

1. $\lim_{t\to 0} \lambda(t)\cdot\tilde{u}$ exists if and only if $\mu(u,\lambda)\leq 0$.
2. $\lim_{t\to 0} \lambda(t)\cdot\tilde{u}=0$ if and only if $\mu(u,\lambda)<0$.
3. $\mu(gx,g\lambda g^{-1})=\mu(x,\lambda)$.

Theorem 3.2.1.10 (Hilbert-Mumford Criterion)**.** *Let G be a linearly reductive group acting on a G-equivariant closed subscheme $U\subseteq\mathbb{P}(V)$, where V is a finite-dimensional representation of G. Let $u\in\mathbb{P}(V)$ be a $\mathbb{C}$-point with a lift $\tilde{u}\in\mathbb{A}(V)$. Then:*

$$\begin{aligned} u\in U^{ss} &\iff 0\notin\overline{G\cdot\tilde{u}} \\ &\iff \lim_{t\to 0}\lambda(t)\cdot\tilde{u}\neq 0 \textit{ for all } \lambda\in \textit{1-PS}(G) \\ &\iff \mu(u,\lambda)\geq 0 \textit{ for all } \lambda\in \textit{1-PS}(G) \end{aligned}$$

In addition, if the action of G has finite stabilizers, then

$$\begin{aligned} u\in U^{s} &\iff G\cdot\tilde{u}\subseteq\mathbb{A}(V) \textit{ is closed} \\ &\iff \mu(u,\lambda)>0 \textit{ for all non-trivial } \lambda\in \textit{1-PS}(G) \end{aligned}$$

Proof. See [Alp26, Theorem 7.4.4] □

3.2.2 Stability Conditions

Definition 3.2.2.1. Let S be a scheme (in our case, $S=\mathrm{Spec}(\mathbb{C})$). A **stable** (resp. **semistable**) curve of genus $g\geq 2$ over S is a proper flat morphism $\pi: C\to S$ such that for all $s\in S$ the fibre C_s of π over s, satisfies:

1. C_s is a reduced, connected, scheme of dimension 1 with $h^1(C_s,\mathcal{O}_{C_s})=g$.
2. Each singular point of C_s is an ordinary double point.
3. If E is an irreducible component of C_s such that $E\cong\mathbb{P}^1$, then E meets the other component of C_s in at least 3 points (resp. 2 points).

Definition 3.2.2.2. Let $C \subset \mathbb{P}^N$ be a complete curve and $\mathcal{L}$ be the restriction of the ample line bundle $\mathcal{O}_{\mathbb{P}^N}(1)$ to C. By Serre's vanishing theorem, there exists $m_0 \in \mathbb{Z}$ such that for all $m > m_0$, $H^i(C, \mathcal{L}^{\otimes m}) = 0$ for all $i > 0$, and $H^0(C, \mathcal{L}^{\otimes m})$ is generated by global sections. Then, the restriction

$$\phi_m : H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m)) \to H^0(C, \mathcal{L}^{\otimes m})$$

is surjective. Assume that $m > m_0$ and take the $P(m)$-th exterior powers. Then, we have the surjection

$$\bigwedge^{P(m)} \phi_m : \bigwedge^{P(m)} H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m)) \twoheadrightarrow \bigwedge^{P(m)} H^0(C, \mathcal{L}^{\otimes m}) \cong \mathbb{C}$$

which defines a point in the projective space $\mathbb{P}(\bigwedge^{P(m)} H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m)))$. We define the m**-th Hilbert point** of the curve C as

$$[H_m(C)] = \bigwedge^{P(m)} \phi_m \in \mathbb{P}(\bigwedge^{P(m)} H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m))).$$

Now, we choose a basis $x_0, x_1, \ldots, x_N$ of $H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(1))$. We have an action of $GL(N+1, \mathbb{C})$ (and an action of $SL(N+1, \mathbb{C})$ since it is a subgroup of $GL(N+1, \mathbb{C})$) on $H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(1))$ by left multiplication. Define that action as

$$[a_{ij}] \cdot x_k = \sum_{l=0}^{N} a_{kl} x_l, \quad a_{ij} \in GL(N+1, \mathbb{C}), \quad (0 \leq k \leq N)$$

We also have an induced $SL(N+1, \mathbb{C})$ action on $\mathbb{P}(\bigwedge^{P(m)} H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m)))$ as follows:

Let $B_m = \{M_1, \ldots, M_\alpha\}$ be the basis of $H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m))$ where M_i's are monomials in degree m in $x_0, \ldots, x_N$ and $\alpha = h^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m))$. The action of $SL(N+1, \mathbb{C})$ on $H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m))$ is given by

$$g \cdot (x_0^{\gamma_0} x_1^{\gamma_1} \cdots x_N^{\gamma_N}) = h_0^{\gamma_0} h_1^{\gamma_1} \cdots h_N^{\gamma_N}$$

where $g \in SL(N+1, \mathbb{C})$, $x_0^{\gamma_0} x_1^{\gamma_1} \cdots x_N^{\gamma_N} \in B_m$ and $h_i = g \cdot x_i$ for $0 \leq i \leq N$. Moreover, we get an action of $SL(N+1, \mathbb{C})$ on $\bigwedge^{P(m)} H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m))$ by

$$g \cdot (M_{i_1} \wedge M_{i_2} \wedge \cdots \wedge M_{i_{P(m)}}) = g \cdot M_{i_1} \wedge g \cdot M_{i_2} \wedge \cdots \wedge g \cdot M_{i_{P(m)}}, \quad M_{i_j} \in B_m$$

(Recall that $M_{i_1} \wedge M_{i_2} \wedge \cdots \wedge M_{i_{P(m)}}$ is a basis for $\bigwedge^{P(m)} H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m))$ with $1 \leq i_1 < i_2 < \cdots < i_{P(m)} \leq \alpha$).

Take the dual action on $\mathbb{P}(\bigwedge^{P(m)} H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m)))^\vee$ which gives an action of $SL(N+1, \mathbb{C})$ on $\mathbb{P}(\bigwedge^{P(m)} H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m)))$.
Let $\lambda \in$ 1-PS$(SL(N+1, \mathbb{C}))$. Take the basis $v_0, v_1, \ldots, v_N$ of $H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(1))$ and $r_0, r_1, \ldots, r_N \in \mathbb{Z}$ such that the action of λ is

$$\lambda(t) \cdot v_i = t^{r_i} v_i, \quad t \in \mathbb{G}_m, \quad 0 \leq i \leq N$$

Let $B'_m = \{M'_1, M'_2, \ldots, M'_\alpha\}$ be a basis of $H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m))$ consisting of monomials with degree m in $v_0, v_1, \ldots, v_N$.

Definition 3.2.2.3. For a monomial $M = v_0^{\gamma_0} v_1^{\gamma_1} \cdots v_N^{\gamma_N}$, we define its λ-weight as $w_\lambda(M) = \sum_{i=0}^N \gamma_i r_i$. The total λ-weight of monomials $M''_1, \ldots, M''_t$ is $\sum_{i=1}^t w_\lambda(M''_i)$. Let β be a set of polynomials and let λ-weight of a polynomial f be the largest weight of a monomial in f. Then, we define $w_\lambda(\beta) = \sum_{f_i \in \beta} w_\lambda(f_i)$

Definition 3.2.2.4 (Hilbert-Mumford index for Hilbert points)**.** Let $C \subseteq \mathbb{P}^N$ be a complete curve and $m \in \mathbb{Z}$ such that $m > m_0$. Then, we have a Hilbert-Mumford index such as

$$\mu_m^{Hilb}(C, \lambda) = \min_\beta w_\lambda(\beta)$$

where the minimum is over all $\beta \subseteq H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(m))$ mapping to a basis of $H^0(C, \mathcal{L}^{\otimes m})$.

Theorem 3.2.2.5. *Let* Hilb^{ss} *be the semi-stable Hilbert locus and* Hilb^s *be the stable Hilbert locus. Then,*

$$C \in \mathrm{Hilb}^{ss} \iff \mu_m^{Hilb}(C, \lambda) \leq 0 \text{ for all } \lambda \in \text{1-PS}(SL(N+1, \mathbb{C}))$$
$$C \in \mathrm{Hilb}^{s} \iff \mu_m^{Hilb}(C, \lambda) < 0 \text{ for all non-trivial } \lambda \in \text{1-PS}(SL(N+1, \mathbb{C}))$$

Proof. This is straightforward from the definition of the Hilbert-Mumford index and the Hilbert-Mumford criterion. □

Theorem 3.2.2.6 (Asymptotic Stability Criterion)**.** *Let $C \subseteq \mathbb{P}^N$ be a nonsingular, nondegenerate curve of genus $g \geq 2$, embedded by the ν-canonical bundle $\omega_C^{\otimes \nu}$ for $\nu \geq 3$. Then, there exists an $m_0 \in \mathbb{Z}$ such that for all $m > m_0$, the Hilbert point $H_m(C)$ is stable.*

Sketch of Proof. The proof relies on applying the Hilbert-Mumford numerical criterion to the m-th Hilbert point of C. For any non-trivial one-parameter subgroup $\lambda \in$ 1-PS$(SL(N+1,\mathbb{C}))$, the action of λ can be diagonalized on the vector space $H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(1))$. This diagonalization naturally induces a weight filtration on $H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(1))$, which we can subsequently restrict to obtain a weight filtration on the homogeneous coordinate ring of the curve C.

To evaluate the Hilbert-Mumford index $\mu_m^{Hilb}(C, \lambda)$, one needs to understand the total weight of a basis of $H^0(C, \mathcal{O}_C(m))$ adapted to this filtration. By relating this λ-weight filtration to the Plücker coordinates of the embedding, the problem translates into bounding the dimensions of the successive quotients of the filtration (or equivalently, bounding the dimension of the linear series spanned by sections of a given weight). These dimensions can be bounded from above using classical results from the theory of algebraic curves, most notably the Riemann-Roch theorem and Clifford's theorem.

When we substitute these dimension bounds back into the weight formula for the Hilbert-Mumford index, the resulting expression is a polynomial in m. A careful analysis shows that the leading coefficient of this polynomial (in terms of m) is strictly negative. This negativity is directly guaranteed by the fact that the curve is embedded via the ν-canonical bundle, provided ν is sufficiently large. Therefore, for all $m \gg 0$, the dominant term dictates that $\mu_m^{Hilb}(C, \lambda) < 0$ for every non-trivial $\lambda \in$ 1-PS$(SL(N+1,\mathbb{C}))$. Concluding by the Hilbert-Mumford criterion, the Hilbert point $H_m(C)$ is stable. □

Remark 3.2.2.7. For a rigorous treatment of the above theorem, we refer the reader to [Gie82, Theorem 1.0.0].

Definition 3.2.2.8. We call a connected curve C of genus g and degree d in $\mathbb{P}^N$ **potentially stable** if it satisfies the following:

1. The embedded curve C is non-degenerate in $\mathbb{P}^N$.
2. The abstract curve C is semistable.
3. The linear series embedding C is complete and nonspecial, i.e., $h^0(C, \mathcal{O}_C(1)) = d + 1 - g$ and $h^1(C, \mathcal{O}_C(1)) = 0$.
4. Any chain of smooth rational components of C meeting the rest of C in exactly two points has length 1.
5. If R is a smooth rational component meeting the rest of C in exactly two points, then $\deg_R(\mathcal{O}_C(1)) = 1$.
6. If Y is a complete subcurve of C with genus $g(Y)$ and degree $\deg(Y)$ meeting the rest of C in p_Y points, then
$$\left|\deg_Y(\mathcal{O}_C(1)) - \frac{d}{g-1}\left(g(Y) - 1 + \frac{p_Y}{2}\right)\right| \leq \frac{p_Y}{2}$$

Theorem 3.2.2.9 (Potential Stability Theorem)**.** *Fix integers $g \geq 2$ and $d \geq 10(g-1)$. Then there exists an $m_0 \in \mathbb{Z}$ such that for all $m > m_0$, if $C \subseteq \mathbb{P}^N$ is a connected curve with a semistable m-th Hilbert point, then C is a potentially stable curve.*

Proof. We will use the contrapositive: if C fails any of the conditions of potential stability, then there exists a $\lambda \in$ 1-PS$(SL(N+1, \mathbb{C}))$ that destabilizes C, i.e., $\mu_m^{Hilb}(C, \lambda) > 0$. We will analyze each condition separately.

1. Assume C is degenerate. Then it is contained in a hyperplane. There exists a non-trivial $\lambda \in$ 1-PS$(SL(N+1, \mathbb{C}))$ such that λ scales the coordinate transverse to this hyperplane with a large negative weight, which destabilizes C.
2. Assume C is not semistable. Then it contains a smooth rational component Y (a rational tail) meeting the rest of C in exactly one point ($p_Y = 1$). Applying

the basic inequality to Y yields a contradiction, because $\deg_Y(\mathcal{O}_C(1)) \geq 1$ (as the embedding is very ample) and d is large, meaning the inequality cannot hold. Thus, the failure of semistability violates the basic inequality, which corresponds to a destabilizing 1-PS.

3. Assume the linear series embedding C is incomplete or nonspecial. If it is incomplete, C lies in a proper linear subspace of $\mathbb{P}^N$, reducing to the first case. If it is special, $h^1(C, \mathcal{O}_C(1)) > 0$, altering the Riemann-Roch dimension count. This discrepancy allows choosing a 1-PS λ whose weights heavily penalize the extra sections, rendering C unstable.

4. Assume there is a chain of smooth rational components of length greater than 1 meeting the rest of C in two points. We can construct a 1-PS that assigns weights to the vertices of the dual graph of C such that the weights strictly decrease along the chain. This concentrated weight distribution destabilizes C.

5. Assume there is a smooth rational component R meeting the rest of C in exactly two points, but $\deg_R(\mathcal{O}_C(1)) \neq 1$. By weighting the sections supported on R differently from the rest of the curve, one calculates that the Hilbert-Mumford index is positive, again destabilizing the curve.

6. Assume the basic inequality fails for some complete subcurve $Y \subset C$. We choose a basis for $H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(1))$ adapted to the subspace of sections vanishing on Y and define a 1-PS λ assigning different weights depending on the order of vanishing on Y. The calculation of the Hilbert-Mumford index for this λ as $m \to \infty$ directly yields the basic inequality; its failure thus implies $\mu_m^{Hilb}(C, \lambda) > 0$.

□

Theorem 3.2.2.10. *Let $\nu \geq 5$. Then there exists a $m \gg 0$ such that every ν-canonically embedded smooth curve is m-Hilbert stable and such that if $C \subseteq \mathbb{P}^N$ is m-Hilbert stable, then C is a ν-canonically embedded stable curve.*

Proof. For the first direction, if $C \subseteq \mathbb{P}^N$ is a ν-canonically embedded smooth curve, then its Hilbert point $H_m(C)$ is well-defined for all sufficiently large m. The condition

$\nu \geq 5$ ensures that the degree $d = \deg(\omega_C^{\otimes\nu}) = 2\nu(g-1)$ satisfies the requirement of the previous theorem ($d \geq 10(g-1)$). By the Asymptotic Stability Criterion, as m grows, the m-th Hilbert point becomes asymptotically stable. Thus, every ν-canonically embedded smooth curve is m-Hilbert stable for sufficiently large m.

Conversely, suppose $C \subseteq \mathbb{P}^N$ is m-Hilbert stable for some $m \gg 0$. The theorem states that such a curve must be a stable curve (no totally degenerate rational components, no non-stable nodal components, etc.). By definition, the linear series embedding C is complete and nonspecial. Let $\mathcal{L} = \mathcal{O}_C(1)$. We know $h^0(C, \mathcal{L}) = d+1-g$. The final, crucial step (due to [Gie82]) is showing that among all line bundles L' on C with $h^0(C, L') = d+1-g$ and $\deg(L') = d$, only $L' = \omega_C^{\otimes\nu}$ yields a stable Hilbert point. Since C is m-Hilbert stable, its embedding bundle $\mathcal{O}_C(1)$ must be exactly $\omega_C^{\otimes\nu}$. Therefore, C is a ν-canonically embedded stable curve. □

3.3 Stable Reduction and Some Deformation Theory

3.3.1 Stable Reduction Theorem

Theorem 3.3.1.1 (Stable Reduction Theorem)**.** *Let R be a discrete valuation ring with quotient field k. Let C be a smooth geometrically irreducible curve over k with genus $g \geq 2$. Then there exists a finite algebraic extension k' of k and a stable curve $C_{k'}$ over R' (integral closure of R) with generic fibre $C_{k',\eta} \cong C \times_k k'$.*

Proof. See [DM69, Corollary 2.7] □

Theorem 3.3.1.2 (Valuative Criterion for Universal Closedness and Properness)**.** *Suppose $f : X \to S$ is a morphism of finite type of locally Noetherian schemes. Then f is universally closed (resp. proper) if and only if for any DVR R with fraction field K and any diagram of the form*

$$\begin{array}{ccc} \operatorname{Spec} K & \longrightarrow & X \\ {\scriptstyle \text{open emb.}}\downarrow & & \downarrow{\scriptstyle f} \\ \operatorname{Spec} R & \longrightarrow & S \end{array}$$

there exists at least one (resp. exactly one) morphism $\operatorname{Spec} R \to X$ *such that the resulting diagram commutes.*

Proof. See [Vak24, Theorem 13.7.6] □

Lemma 3.3.1.3 (Valuative Criterion for Universal Closedness for Algebraic Stacks)**.** *Let* $f : \mathcal{X} \to \mathcal{Y}$ *be a morphism of stacks. If*

- *(i)* f *is quasi-compact*
- *(ii)* f *satisfies the existence part of the valuative criterion (Theorem 3.3.1.2)*

Then f *is universally closed.*

Proof. See [Sta26, Lemma 101.42.1] □

3.3.2 Deformation Theory

In this section, we will state the necessary definitions and theorems of deformation theory for constructing the moduli stack $\overline{\mathcal{H}}_g$. The rigorous treatment of the topic is beyond the scope of this thesis, but these results are essential for the construction of the moduli stack. The reader is encouraged to consult [Ser06] and [Har10].

Throughout this section we will use the following notation:

- $\mathcal{A}$ is the category of local artinian k-algebras with residue field k.
- $\mathcal{A}^*$ is the category of local noetherian k-algebras with residue field k.
- $\widehat{\mathcal{A}}$ is the category of complete local noetherian k-algebras with residue field k.
- $\mathcal{A}_\Lambda$ is the category of local artinian Λ-algebras with residue field k, where $\Lambda \in ob(\mathcal{A}^*)$.

Definition 3.3.2.1. Let $A \to R$ be a ring homomorphism. An **extension of A-algebras** (or A-extension of R) is a short exact sequence

$$(R', \phi) : 0 \to I \to R' \xrightarrow{\phi} R \to 0$$

where R' is an A-algebra and ϕ is an A-algebra homomorphism such that $\ker(\phi) = I$ and $I^2 = (0)$.

Definition 3.3.2.2. For every A-algebra R and for every R-module I, the $\mathrm{Ex}_A(R, I)$ is the set of isomorphism classes of A-extensions of R by I.

Definition 3.3.2.3. Let X be an algebraic scheme. A cartesian diagram of morphisms of schemes

$$\eta: \quad \begin{array}{ccc} X & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow{\scriptstyle \pi} \\ \mathrm{Spec}(k) & \xrightarrow{s} & S \end{array}$$

where π is surjective and flat, and S is connected, is called a **family of deformations** (or deformation) of X parametrized by S. We call S the **parameter scheme** and $\mathcal{X}$ the **total scheme**. If S is an algebraic scheme, then the fiber $\mathcal{X}(t)$ is also called a deformation of X for every k-point $t \in S$.
When $S = \mathrm{Spec}(A)$ is affine, where $A \in ob(\mathcal{A}^*)$ and $s \in S$ is the closed point, we have a **local deformation**.
The local deformation $(\mathrm{Spec}(A), \eta)$ is said to be **infinitesimal** if $A \in ob(\mathcal{A})$.
The local deformation $(\mathrm{Spec}(A), \eta)$ is said to be **first-order** if $A = k[\epsilon]$ (ring of dual numbers).

Let X be a nonsingular algebraic variety. Then a **small extension** is the following short exact sequence of local artinian k-algebras

$$e: 0 \to (t) \to \tilde{A} \to A \to 0$$

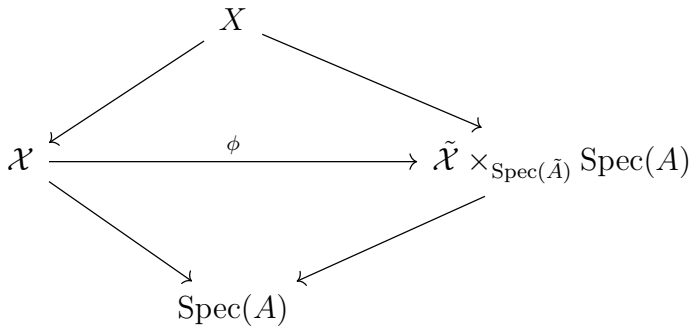
and let

$$\xi: \quad \begin{array}{ccc} X & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(A) \end{array}$$

be an infinitesimal deformation of X. A *lifting* of ξ to $\tilde{A}$ consists in a deformation

$$\tilde{\xi}: \quad \begin{array}{ccc} X & \longrightarrow & \tilde{\mathcal{X}} \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \longrightarrow & \mathrm{Spec}(\tilde{A}) \end{array}$$

and an isomorphism of deformations



Proposition 3.3.2.4. *Let $A \in ob(\mathcal{A})$ and let ξ be a an infinitesimal deformation of X over A. Then for every small extension, there exist an associated element $o_\xi(e) \in H^2(X, T_X)$ called the **obstruction** to lifting ξ to $\tilde{A}$. $o_\xi(e) = 0$ if and only if a lifting of ξ to $\tilde{A}$ exists.*

Proof. See [Ser06, Proposition 1.2.12]. □

Definition 3.3.2.5. Let F be a functor of Artin rings. Suppose that $v(F)$ is a k-vector space such that for every $A \in ob(\mathcal{A}_\Lambda)$ and every $\xi \in F(A)$, there exists a linear map

$$\xi_v : Ex_\Lambda(A, k) \to v(F)$$

$\ker(\xi_v)$ is the isomorphism classes of extensions $(\tilde{A}, \phi)$ such that

$$\xi \in Im(F(\tilde{A}) \to F(A))$$

We call $v(F)$ an **obstruction space** of F. If $v(F) = (0)$, it is called **unobstructed**.

Theorem 3.3.2.6. *F has (0) as an obstruction space if and only if it is smooth.*

Proof. See [Ser06, Proposition 2.2.10]. □

Let X be an algebraic scheme. Then for any $A \in ob(\mathcal{A})$ we define

$$\mathrm{Def}_X(A) = \{\text{deformations of } X \text{ over } A\}/\text{isomorphism}$$

This defines a functor of Artin rings as

$$\mathrm{Def}_X : \mathcal{A} \to Sets$$

which is called the **local moduli functor**.

Theorem 3.3.2.7. *Let X be an algebraic scheme. Then we have a natural identification*

$$\mathrm{Def}_X(k[\epsilon]) = \mathrm{Ex}_k(X, \mathcal{O}_X)$$

and a short exact sequence

$$0 \to H^1(X, T_X) \to \mathrm{Def}_X(k[\epsilon]) \to H^0(X, T_X^1) \to H^2(X, T_X) \to 0 \tag{3.2}$$

Moreover, if X is reduced, then there is an isomorphism

$$\mathrm{Def}_X(k[\epsilon]) \cong \mathrm{Ext}^1_{\mathcal{O}_X}(\Omega^1_X, \mathcal{O}_X)$$

and the short exact sequence (3.2) *is isomorphic to*

$$0 \to H^1(X, T_X) \to \mathrm{Ext}^1_{\mathcal{O}_X}(\Omega^1_X, \mathcal{O}_X) \to H^0(X, \mathcal{E}xt^1_{\mathcal{O}_X}(\Omega^1_X, \mathcal{O}_X)) \to H^2(X, T_X)$$

Definition 3.3.2.8. Let k be a field (not necessarily algebraically closed). Let S be a finite type k-algebra.

1. We say that S is a **global complete intersection** if there exists a presentation $S = k[x_1, ..., x_n]/(f_1, ..., f_r)$ such that $\dim(S) = n - r$.
2. We say S is a **local complete intersection** (l.c.i) if there exists a covering $\mathrm{Spec}(S) = \bigcup D(g_i)$ such that each S_{g_i} is a global complete intersection.

Theorem 3.3.2.9. *Let C be a reduced l.c.i algebraic scheme over a field k with $char k = 0$. Then $\mathrm{Ext}^2_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C)$ is an obstruction space for the functor Def_C.*

Proof. See [Ser06, Proposition 2.4.8]. □

Example 3.3.2.10. Let C be a smooth algebraic curve over a field k. Then C is reduced and l.c.i. Then $\mathcal{E}xt^1_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C)$ is a torsion sheaf and $\mathcal{E}xt^2_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C) = (0)$. Then

- $H^2(C, \mathcal{H}om(\Omega^1_C, \mathcal{O}_C)) = (0)$ since C is a curve.
- $H^1(C, \mathcal{E}xt^1_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C)) = (0)$ since $\mathcal{E}xt^1_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C)$ is a torsion sheaf and first cohomology of torsion sheaves is zero.
- $H^0(C, \mathcal{E}xt^2_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C)) = (0)$ since $\mathcal{E}xt^2_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C) = (0)$.

By Theorem 3.3.2.9, C is unobstructed.

CHAPTER 4

THE MODULI STACKS OF HYPERELLIPTIC CURVES

4.1 Construction of the GIT Quotient

We fix the genus $g \geq 2$, the integer $\nu \geq 3$, the Hilbert polynomial $P(m) = (2m\nu - 1)(g-1)$, integer $N = (2\nu - 1)(g-1) - 1$, and our base field $k = \mathbb{C}$.

Firstly, we define $\mathcal{H}^{ss} = \mathrm{Hilb}^{ss}$ be the semistable locus of $\mathcal{H} = \mathrm{Hilb}_N^{P(m)}$ under the action of $PGL(N+1, \mathbb{C})$ and $\mathcal{H}^s = \mathrm{Hilb}^s$ be the stable locus of $\mathcal{H} = \mathrm{Hilb}_N^{P(m)}$. Define K to be the locus in $\mathcal{H}$ of stable curves C with genus g and embedded into $\mathbb{P}^N$ by the ν-th tensor power of their canonical bundle ω_C. Finally, define $K^{ss} = K \cap \mathrm{Hilb}^{ss}$ and $K^s = K \cap \mathrm{Hilb}^s$.

Theorem 4.1.0.1. *(1) K^{ss} is smooth.*

(2) K^{ss} is closed in Hilb^{ss}.

(3) $K^{ss} = K^s$.

(4) K^{ss} contains ν-canonical embedded models of every stable curve of genus g.

Proof of (1). The $\mathcal{H}^{ss}$ is an open subscheme of $\mathcal{H}$. By definition K^{ss} is open in K. Thus, we need to prove that K is formally smooth. To do that we need to check the deformations of K.

Let $\mathrm{Def}(C)$ denote the deformation space of the curve C. Now, take a closed point $[C] \in K$ which corresponds to a stable curve C and its embedding via $\omega_C^{\otimes \nu}$. By definition, C is a curve with at worst nodal singularities, so it is an l.c.i. scheme.

By Theorem 3.3.2.9 its obstruction space is given by $\mathrm{Ext}^2_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C)$. By Theorem 3.3.2.9 and Example 3.3.2.10, $\mathrm{Ext}^2_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C) = 0$. Thus, local deformation space $\mathrm{Def}(C)$ is formally smooth by Theorem 3.3.2.6. By definition, points of K are ν-canonically embedded stable curves. Choosing a basis for the $H^0(C, \omega_C^{\otimes\nu})$ corresponds to a free and transitive action of $PGL(N+1, \mathbb{C})$. Thus, the forgetful map $K \to \mathrm{Def}(C)$ makes K a $PGL(N+1, \mathbb{C})$-torsor. A torsor over a smooth base is a smooth scheme.

Therefore, K is smooth and thus K^{ss} is smooth. □

Proof of (2). Since $\mathcal{H}^{ss}$ is locally of finite type over $\mathbb{C}$, it is locally Noetherian. Now, let R be a DVR containing $\mathbb{C}$ and let $A = \mathrm{Frac}(R)$ be its fraction field.

Let $\Delta = \mathrm{Spec}(A)$ and $\eta = \mathrm{Spec}(K)$ be the generic point of Δ and closed point $\{0\} = \mathrm{Spec}(\mathbb{C})$. Assume we have a morphism $\Delta \to \mathcal{H}^{ss}$ such that the η maps into K^{ss}. This generic point corresponds to a ν-canonically embedded curve $C_\eta \hookrightarrow \mathbb{P}^N_A$. Let Z_0 be the central limit of family.

By Theorem 3.3.1.1, C_η is potentially stable. Furthermore, there exists a finite field extension A' of A. Now, let R' be the integral closure of R in A'. Since $\mathbb{C}$ is algebraically closed, the residue field of R' is still $\mathbb{C}$. Over the base $\mathrm{Spec}(R')$, we have a base-changed curve $C_{\eta'} = C_\eta \times_A A'$ which extends to a unique family of flat, proper stable curves $\mathcal{C}' \to \mathrm{Spec}(R')$. Let C'_0 be the central fiber of this new family of curves. Because the residue field is still $\mathbb{C}$, C'_0 is a stable curve over $\mathbb{C}$.

Let $\omega^{\otimes\nu}_{\mathcal{C}'/R'}$ be the ν-canonical bundle on $\mathcal{C}'$. Because its fibers are stable and $\nu \geq 3$, $\omega^{\otimes\nu}_{\mathcal{C}'/R'}$ is very ample. Thus, it defines an embedding $\mathcal{C}' \hookrightarrow \mathbb{P}^N_{R'}$. This embedding gives us a new morphism of schemes $\mathrm{Spec}(R') \to \mathcal{H}$. Since the central fiber is a stable curve embedded by its ν-canonical bundle, its corresponding point is an semistable m-Hilbert point denoted as $Z'_0 \in K^{ss}$ by Theorem 3.2.2.10.

Since the residue fields do not change, we are comparing two $\mathbb{C}$-points in the Hilbert scheme, Z_0 and Z_0'. Over the generic point of $\mathrm{Spec}(R')$ they represent the same abstract curve differing by a choice of coordinates in $\mathbb{P}^N$. Thus, they lie on the same $PGL(N+1,\mathbb{C})$-orbit. By the definition of GIT quotient $\mathcal{H}^{ss} \to \mathcal{H}^{ss}//PGL(N+1,\mathbb{C})$ is separated. We know that limits of orbits must intersect in a separated quotient. Also we know that stable points have closed orbits. Thus, Z_0 and Z_0' $PGL(N+1,\mathbb{C})$-equivalent. Since $Z_0' \in K^{ss}$, then $Z_0 \in K^{ss}$. Therefore, K^{ss} is closed in $\mathcal{H}^{ss}$.

$$[Z_0] = [Z_0'] \in \mathcal{H}^{ss}/PGL(N+1,\mathbb{C}).$$

$$[Z_0] \in K^{ss}/PGL(N+1,\mathbb{C}).$$

Thus, $K^{ss}/PGL(N+1,\mathbb{C})$ is closed in $\mathcal{H}^{ss}/PGL(N+1,\mathbb{C})$. □

Proof of (3). By definition of stable and semistable loci, we know that $K^s \subset K^{ss}$. Now, we need to show that $K^{ss} \subset K^s$. Take a point $[C] \in K^{ss}$ which corresponds to a ν-canonically embedded semistable curve $C \subset \mathbb{P}^N$. Since $\nu \geq 5$, by Theorem 3.2.2.10, there exists an $m \gg 0$ such that m-th Hilbert point of C, $H_m(C)$ is in $\mathcal{H}^s$. By definition of $K^s = \mathcal{H}^s \cap K$, $[C] \in K^s$.

Therefore, $K^{ss} \subset K^s$ and thus $K^{ss} = K^s$. □

Proof of (4). Let C be a stable curve of genus g and $\omega_C^{\otimes\nu}$ define an embedding of C into $\mathbb{P}^N$. Then again by Theorem 3.2.2.10, there exists a point $[C]$ in $\mathcal{H}^s$ which corresponds to a ν-canonically embedded C.

Therefore, $[C] \in K^s = K^{ss}$ and thus K^{ss} contains a model of every stable curve of genus g embedded by its ν-canonical bundle. □

Now, we need to define the hyperelliptic locus in K. Define K°_{hyp} to be the subscheme of K consisting of ν-canonically embedded hyperelliptic curves of genus g. Set-theoretically, it contains smooth curves C which admits a base-point free linear series of degree 2 and dimension 1, i.e. g^1_2. Let $\overline{K^\circ_{hyp}}$ be the Zariski closure.

Theorem 4.1.0.2. *(1)* $K^{ss}_{hyp} = \overline{K^{\circ}_{hyp}} \cap \mathcal{H}^s$ *parametrizes admissible hyperelliptic curves.*

(2) K^{ss}_{hyp} *is closed in* $\mathcal{H}^{ss}$*.*

(3) K^{ss}_{hyp} *is smooth.*

(4) $K^{ss}_{hyp} = K^{s}_{hyp}$

(5) K^{ss}_{hyp} *contains* ν*-canonical embedded models of every smooth hyperelliptic curve.*

Proof of (1). Let $\Delta = \mathrm{Spec}(\mathbb{C}[[t]])$ with generic point η and closed point 0. Assume we have a family of curves $\mathcal{C} \to \Delta$ that maps into K^{ss} where the image of the generic fiber C_η lies in K°_{hyp}. Since C_η is a smooth hyperelliptic curve, it has a unique hyperelliptic involution $\iota_\eta \in Aut(C_\eta)$ with order 2. Let $\mathrm{Aut}(\mathcal{C}/\Delta)$ be the automorphism group of the family.

Then $\mathrm{Aut}(\mathcal{C}/\Delta)$ is a finite, unramified group scheme over Δ. Recall that finite morphisms are proper. Then, by Theorem 3.3.1.2, we get a unique global involution $\iota \in \mathrm{Aut}(\mathcal{C})$ which lifts ι_η.

Now let C_0 be the central fiber and then we have ι_0 which is the restriction of ι to C_0. Let $\mathcal{T} = \mathcal{C}/\langle\iota\rangle \to \Delta$ be a new proper family of curves. Then, over the generic point η, we have $T_\eta = C_\eta/\langle\iota_\eta\rangle \cong \mathbb{P}^1$. We know that $\mathbb{P}^1$ has genus 0. Since the genus of fibers is constant in flat, proper families, $T_0 = C_0/\langle\iota_0\rangle$ has genus 0. We know that C_0 is a stable nodal curve, and its quotient with a finite group ι_0 is also a nodal curve. Thus, T_0 is a nodal curve with genus 0.

The map $f_0 : C_0 \to T_0$ has degree 2 since ι_0 has order 2. Take any node $p \in C_0$, the local ring $\mathcal{O}_{C_0,p} \cong \mathbb{C}[[x,y]]/(xy)$. We know that hyperelliptic involutions must preserve nodes. Since ι_0 has order 2, its action on coordinates is either $(x,y) \mapsto (y,x)$ or $(x,y) \mapsto (-x,-y)$. In both cases, nodes of C_0 maps to nodes of T_0. Thus, f_0 is a degree 2 admissible cover. So, limits of smooth hyperelliptic curves are admissible

hyperelliptic curves.

Now, we need to prove the converse. Assume C_0 is a stable nodal curve that admits a degree 2 cover f_0. We need to show that C_0 is in the K^{ss}_{hyp}. Let T_0 be a nodal curve of genus 0 (a tree of projective lines). By the deformation theory of curves, any tree of projective lines can be smoothed. So, there exists a proper, flat family of curves $\mathcal{T} \to \Delta$ with central fiber T_0 and generic fiber $T_\eta \cong \mathbb{P}^1$. By the deformation theory of admissible covers, the map $f_0 : C_0 \to T_0$ can be deformed along the target. So, there exists a family of curves $\mathcal{C} \to \Delta$ and an extension of f_0 to a degree 2 map $\mathcal{C} \to \mathcal{T}$. The generic fiber of this deformed cover $f_\eta : C_\eta \to T_\eta \cong \mathbb{P}^1$ is a degree 2 map. Since admissible covers preserve the stable reduction property, C_η is smooth and $C_\eta \in K^\circ_{hyp}$. Furthermore, C_0 is the central limit of that family of curves and K^{ss}_{hyp} is the Zariski closure of K°_{hyp}, we have $[C_0] \in K^{ss}_{hyp}$. Thus, admissible hyperelliptic curves are limits of smooth hyperelliptic curves.

Therefore, K^{ss}_{hyp} parametrizes admissible hyperelliptic curves. □

Proof of (2). By definition K^{ss}_{hyp} is closed in K^{ss}. Since K^{ss} is closed in $\mathcal{H}^{ss}$, then K^{ss}_{hyp} is closed in $\mathcal{H}^{ss}$. □

Proof of (3). Take a closed point $x \in K^{ss}_{hyp}$ which corresponds to a ν-canonically embedded stable hyperelliptic curve $C \hookrightarrow \mathbb{P}^N$. From the previous theorem, we know that K^{ss} is smooth. Since $x \in K^{ss}$, its local deformation space is unobstructed. Since C is a hyperelliptic curve, we have a unique involution $\iota \in \mathrm{Aut}(C)$ with order 2. This involution acts naturally on C, which induces a canonical pullback action on the cotangent sheaf and the structure sheaf. By the functoriality, this induces an action on the long exact sequence of Ext groups. Then we get

$$\mathrm{Obs}(K^{ss}_{hyp}) = \mathrm{Ext}^2_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C)^\iota$$

Thus, we are checking ι-invariant elements of $\mathrm{Ext}^2_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C)$. However, $\mathrm{Ext}^2_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C) = 0$ by the previous theorem, so $\mathrm{Ext}^2_{\mathcal{O}_C}(\Omega^1_C, \mathcal{O}_C)^\iota = 0$. Since the obstruction space is 0, the local deformation space of x is unobstructed. Thus, every

first-order deformation of C lifts to a hyperelliptic deformation.

Therefore, K^{ss}_{hyp} is smooth. □

Proof of (4). This follows from the previous theorem and definition of K^{s}_{hyp} and K^{ss}_{hyp}. □

Proof of (5). This follows from the previous theorem and definition of K^{ss}_{hyp}. □

4.2 Moduli Stacks

Theorem 4.2.0.1. *The quotient stack $[K^{ss}_{hyp}/PGL(N+1,\mathbb{C})]$ is a smooth, separated, proper Deligne-Mumford stack.*

Proof. By Theorem 3.2.1.6, we know that $[K^{ss}_{hyp}/PGL(N+1,\mathbb{C})]$ is a geometric quotient. By Theorem 2.2.3.18, it is an algebraic (Artin) stack. Since $K^{ss}_{hyp} = K^{s}_{hyp}$, every point $x \in K^{ss}_{hyp}$ is stable by definition. There exists a stable hyperelliptic curve C_x and $Stab(x) \cong Aut(C_x)$. We know that the automorphism group of a stable curve is finite. So, $Stab(x)$ is finite for every $x \in K^{ss}_{hyp}$. By Theorem 2.2.3.17, $K^{ss}_{hyp}/PGL(N+1,\mathbb{C})$ is a DM-stack.

For smoothness, we have the quotient map $\pi : K^{ss}_{hyp} \to [K^{ss}_{hyp}/PGL(N+1,\mathbb{C})]$, which is a faithfully flat and smooth morphism by Theorem 2.2.3.18. Since $PGL(N+1,\mathbb{C})$ is a smooth algebraic group, the smoothness of the quotient stack depends on the smoothness of K^{ss}_{hyp}. We know that K^{ss}_{hyp} is smooth by the previous theorem. Thus, $[K^{ss}_{hyp}/PGL(N+1,\mathbb{C})]$ is smooth.

For separatedness, since the action of $PGL(N+1,\mathbb{C})$ on K has finite stabilizers, the action of $PGL(N+1,\mathbb{C})$ on $K^{ss}_{hyp} = K^{s}_{hyp}$ is proper and the quotient stack $[K^{ss}_{hyp}/PGL(N+1,\mathbb{C})]$ is separated by Theorem 3.2.1.6.

For properness, a stack of finite type over $\mathrm{Spec}(\mathbb{C})$ is proper if it is separated and universally closed. We know that the quotient stack is separated. For universal closedness, we have already proved that $K^s_{hyp} = K^{ss}_{hyp}$ is universally closed by Theorem 3.3.1.2 and Theorem 3.3.1.1. We can extend that to our stack via the composition $\mathrm{Spec}(R') \to K^{ss}_{hyp} \to [K^{ss}_{hyp}/PGL(N+1,\mathbb{C})]$, and apply Theorem 3.3.1.3

Therefore, $[K^{ss}_{hyp}/PGL(N+1,\mathbb{C})]$ is a smooth, separated, proper Deligne-Mumford stack. □

Remark 4.2.0.2. In the literature, the construction of $\overline{\mathcal{H}_g}$ (and $\overline{\mathcal{M}_g}$) is often referred to as the *Deligne-Mumford compactification*. In general topology, a compact space is required to be Hausdorff, but in algebraic geometry, spaces with the Zariski topology are rarely Hausdorff. Therefore, the algebraic analogue of a "compact Hausdorff" space is a **proper** space (separated and universally closed). Because we are working over $\mathrm{Spec}(\mathbb{C})$, Serre's GAGA theorems [Ser56] guarantee that the properness of the algebraic stack $\overline{\mathcal{H}_g}$ is equivalent to its underlying analytic space being compact and Hausdorff in the usual Euclidean topology. Thus, the terminology "compactification" is fully justified.

Theorem 4.2.0.3. *The Deligne-Mumford stack* $[K^{ss}_{hyp}/PGL(N+1,\mathbb{C})]$ *is the moduli space* $\overline{\mathcal{H}_g}$*, i.e.,* $[K^{ss}_{hyp}/PGL(N+1,\mathbb{C})] \cong \overline{\mathcal{H}_g}$.

Proof. First, let us define a natural transformation Θ as:

$$\Theta : [K^{ss}_{hyp}//PGL(N+1,\mathbb{C})] \to \overline{\mathcal{H}_g}$$

By definition of Hilbert schemes, it has a universal family of curves. If we restrict this family to K^{ss}_{hyp}, we get a flat, proper universal family of stable hyperelliptic curves that are embedded to $\mathbb{P}^N$ by their ν-canonical bundle $\omega_C^{\otimes\nu}$. So we have;

$$\phi : \mathcal{C}_{hyp} \to K^{ss}_{hyp}$$

The moduli functor (or stack) $\overline{\mathcal{H}_g} : Sch_{\mathcal{C}} \to Groupoids$ is defined as the category fibered in groupoids over $Sch_{\mathbb{C}}$ with its objects are the flat families of stable hyperelliptic curves. Thus, by the universal property of $\overline{\mathcal{H}_g}$ we get a canonical morphism

from K^{ss}_{hyp} to $\overline{\mathcal{H}_g}$.

The $PGL(N+1,\mathbb{C})$ acts on K^{ss}_{hyp} by changing the basis of $\mathbb{P}^N$. This action changes the embeddings but preserves the isomorphism class of the fibers. Since fibers are invariant under the group action, the classyfing morphism is $PGL(N+1,\mathbb{C})$-invarinat and descends to the morphism Θ.

Now, we will show that Θ is essentially surjective. Let C be a stable hyperelliptic curve over $\mathbb{C}$. Thus, it represents a point in $\overline{\mathcal{H}_g}$. Since C is a stable hyperelliptic curve, its line bundle $\omega_C^{\otimes\nu}$ is very ample for $\nu \geq 3$. Thus, C has an embedding to $\mathbb{P}^N$ and since it is stable, it has a representative in K^{ss}_{hyp}. Since C is arbitrary, Θ is essentially surjective.

Lastly, we need to show that Θ is fully faithfull. Let $c_1, c_2 \in K^{ss}_{hyp}$ be the representatives of two embedded curves $C_1, C_2 \subset \mathbb{P}^N$ by their ν-canonical bundle $\omega_{C_i}^{\otimes\nu}$.

First, if c_1 and c_2 are in the same orbit, there exists a $g \in PGL(N+1,\mathbb{C})$ s.t. $g \cdot c_1 = c_2$, i.e., mapping C_1 to C_2. This g restricts to a scheme isomorphism $C_1 \xrightarrow{\sim} C_2$.

Conversely, if we have an isomorphism $\phi : C_1 \xrightarrow{\sim} C_2$ and since both curves are embedded by their ν-canonical bundle, we have $\mathcal{O}_C(1) = \omega_C^{\otimes\nu}$. Thus, the isomorphism ϕ pulls back the dualizing sheaf:

$$\phi^*(\omega_{C_2}^{\otimes\nu}) = (\omega_{C_1}^{\otimes\nu})$$

This pullback induces an isomorphism between the vector spaces of global sections:

$$\phi^* : H^0(C_2, \omega_{C_2}^{\otimes\nu}) \xrightarrow{\sim} H^0(C_1, \omega_{C_1}^{\otimes\nu})$$

This isomorphism corresponds to a change of basis in $\mathbb{P}^N$, which is an element $g \in PGL(N+1,\mathbb{C})$. Thus, C_1 and C_2 are isomorphic if and only if their representatives in K^{ss}_{hyp} are in the same orbit.

Therefore, since Θ is fully faithful and essentially surjective, it is an equivalence of groupoids. Thus, it is an isomorphism of stacks. $\square$

CHAPTER 5

CONCLUSIONS

In this thesis, we surveyed the construction of the moduli stack of stable curves of genus g, denoted by $\overline{\mathcal{M}}_g$, and its restriction to the hyperelliptic locus, $\overline{\mathcal{H}}_g$ via Geometric Invariant Theory (GIT). We realized the stable curves as ν-canonically embedded points in the Hilberts scheme, and we were able to use projective geometry to construct the parameter space. We then isolated the the hyperelliptic locus inside the Hilbert scheme via Brill-Noether theory and used Stable Reduction Theorem and the valuative criterion for universal closedness to show that the hyperelliptic locus is closed subscheme of Hilbert scheme. Finally, we verify that the GIT quotient has finite and reduced stabilizers, so we confirmed that it is indeed a Deligne-Mumford stack. We finished the thesis by showing that DM-stack is equivalent to the moduli functor of stable hyperelliptic curves.

This GIT approach gives us a highly algebraic foundation for the construction of the moduli stack of hyperelliptic curves. However, it is not the only approach to construct it. Another powerful approach is the theory of admissible covers and Hurwitz stacks. This approach uses the fact that hyperelliptic curves are 2-to-1 covers of $\mathbb{P}^1$ rather than embedding them into projective space. It studies the covers and classifies them up to a suitable equivalance relation.

This framework uses Hurwitz stacks to compactify the moduli stack, which uses parametrization of degree 2 admissible covers of genus zero (i.e. $\mathbb{P}^1$) stable curves. As a smooth hyperelliptic curve degenerates into a nodal stable curve, which is a limit point of the stable locus, the hyperelliptic involution and the degree 2 maps are

forced to degenerate into an admissible map to a nodal rational target. The Hurwitz stack method gives a combinatorial description of the boundary divisors of the moduli of hyperelliptic curves by analyzing the distribution of their ramification points on the target curve. For future questions about intersection theory and local deformation theory, the admissible cover approach can be a more suitable framework to work with.

Ultimately, this construction is not an endpoint but a foundation for modern mathematical research. The stack theory and deformation theory detailed in this thesis serve as a building block for the characteristic-p fields and arithmetic geometry. Also, this classical stack construction can be extended to the derived algebraic geometry with derived stacks. This way we can expand our understanding with derived moduli stacks and derived deformation theory.

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